# Jamf 350 Course

## Student Workbook

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# Reference

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# Environment

## Resources

Links to external resources for this course can be found here:
`https://docs.jamf.com/education-services/resources`

The Student Resources directory can be found in the following path on the provided computer:
`/Users/Shared/Resources/`

## Virtual Machines

Virtual machines can be opened in Fusion.app which is installed on the provided computer.

| Name | IP Address | Hostname | Functions |
|---|---|---|---|
| Ubuntu VM 1 | `192.168.56.101` | `vm1.jamf.training` | • database<br>• web application<br>• memcache |
| Ubuntu VM 2 | `192.168.56.102` | `vm2.jamf.training` | • database<br>• web application<br>• memcache |
| Windows VM 3 | `192.168.56.103` | `vm3.jamf.training` | • database<br>• web application |
| Windows VM 4 | `192.168.56.104` | `vm4.jamf.training` | • database<br>• web application |
| Ubuntu VM 5 | `192.168.56.105` | `pro.jamf.training` | • load balancer<br>• memcache<br>• monitoring |

## MySQL Configuration

Enter information created during installation and configuration lessons in the table below.

| L | Name | Database | Account Name | Password |
|---|---|---|---|---|
| 1 | Windows VM 3 | `mysql` | `root@localhost` | |
| 1 | Windows VM 3 | `jamfdb` | `jamf@localhost` | `Jamf1234` |
| 5 | Ubuntu VM 1 | `mysql` | `root@localhost` | |
| 5 | Ubuntu VM 1 | `jamfcluster` | `jamf@localhost` | `jamf101` |
| 10 | Windows VM 3 | | | |
| 10 | Ubuntu VM 1 | `testdb` | `test@localhost` | `test101` |
| 13 | Ubuntu VM 2 | `jamfcluster` | | |
| 13 | Windows VM 4 | `jamfcluster` | | |


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## Jamf Pro Setup

Enter Jamf Pro User names and passwords created with the Setup Assistant in the table below.

| L | IP / Fully Qualified Domain Name (FQDN) | Username | Password |
|---|---|---|---|
| 2 | `192.168.56.103`<br>`https://vm3.jamf.training:8443/` | jamfadmin | Jamf1234 |
| 7 | `192.168.56.101`<br>`https://vm1.jamf.training:8443/` | | |
| 11 | `192.168.56.103`<br>`https://vm3.jamf.training:8443/_____` | | |
| 11 | `192.168.56.101`<br>`http://vm1.jamf.training:9006/testing` | | |
| 16 | `192.168.56.102`<br>`https://vm2.jamf.training:8443` | | |
| 16 | `192.168.56.104`<br>`https://vm4.jamf.training:8443` | | |

## Virtual Machine Credentials

| | Username | Password |
|---|---|---|
| **Mac** | `serveradmin` | `Jamf1234` |
| **Ubuntu** | `serveradmin` | `Jamf1234` |
| **Windows** | `Administrator` | `Jamf1234` |

## Tomcat Manager Credentials

Enter the username and password created for Tomcat Manager during the Windows Tomcat installation in the table below.

| | |
|---|---|
| **Username** | |
| **Password** | |

## Certificate Keystore Password

Enter the password used for the certificate keystore in Java `keytool` in the table below.

| | |
|---|---|
| **Password** | |

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# Ports and Services

Ports and services used by required Jamf Pro components:

**Apache Tomcat -**

| Protocol | Port | Function | Verification |
|---|---|---|---|
| `http` (default) | `8080`<br>`9006` (macOS)<br>`80` (default) | • client traffic<br>• web application | • Tomcat processes<br>• Tomcat logs<br>• Browse to `URL:port`<br>• `curl` to `URL:port –` `/healthCheck.html` |
| `https` | `8443`<br>`443` (default) | • client traffic<br>• web application | • Enable SSL certificate<br>• Tomcat processes<br>• Tomcat logs<br>• Browse to `URL:port`<br>• `curl` to `URL:port –` `/healthCheck.html` |

**Java -**

| Protocol | Port | Function | Verification |
|---|---|---|---|
| JVM / JAVA | defined<br>`3333` (Linux)<br>`8008` (Windows) | webapp / JDBC<br>JSP servlets<br>JMX monitoring | • Java processes<br>• defined<br>• mbeans |

**MySQL -**

| Protocol | Port | Function | Verification |
|---|---|---|---|
| `mysql` | `3306` (default) | database | • MySQL processes<br>• `show processlist;`<br>• MySQL accounts<br>• database backup |


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External ports and services:

| Service | Port | Function | Verification |
|---|---|---|---|
| APNs | 2195<br>2196 | • Push Notifications<br>• feedback | • `telnet gateway.push.apple.com 2195`<br>• `netstat`<br>• Push Diagnostics.app |
| DEP<br>VPP | 443 | • Device Enrollment Program<br>• Volume Purchase Program | • refresh connection in Jamf Pro server<br>• `https://www.apple.com/support/systemstatus/` |
| MDM | 5223 | Mobile Device Management | • *e.g.*, `telnet 5-courier.sandbox.push.apple.com 5223`<br>• `netstat`<br>• Push Diagnostics.app |

Ports and services which can be integrated with Jamf Pro:

| Service | Port | Function |
|---|---|---|
| Lightweight Directory Access Protocol (LDAP) | 389 | Directory |
| LDAP over SSL (LDAPS) | 636 | Directory |
| Apple File Protocol (AFP) | 548 | File Sharing |
| Server Message Block (SMBX) | 137–139<br>445 | File Sharing |
| Simple Mail Transfer Protocol (SMTP) | 25 | Mail |
| Simple Mail Transfer Protocol (SMTPS) | 587 | Mail |
| Secure Shell (SSH) | 22 | Remote Command Execution |
| Screen Sharing (VNC) | 5900 | Remote Access |
| Syslog | 514 | Remote Logging |
| Global Service Exchange (GSX) | 80 | Warranty / Purchase Information |

**NOTE:** Jamf Pro integrations with LDAP, MySQL & email may require inbound port connections from a clustered web application hosted on a perimeter network (DMZ).

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# File Paths

**MySQL**

| | |
|---|---|
| Home Directory on Ubuntu: | `/var/lib/mysql/` |
| MySQL configuration file on Ubuntu: | `/etc/mysql/mysql.conf.d/mysqld.cnf > /etc/mysql/my.cnf` |
| Home Directory on Windows: | `C:\Program Files\MySQL\MySQL Server #.#\` |
| MySQL configuration file on Windows: | `C:\ProgramData\MySQL\MySQL Server #.#\my.ini` |

**Java JDK**

| | |
|---|---|
| Home Directory on Windows: | `C:\Program Files\Java\jdk#.#.#_###\` |
| Java Cryptography Extension (JCE) Location on Windows: | `C:\Program Files\Java\jre#.#.#_###\lib\security\...` |

**Tomcat**

| | |
|---|---|
| Home directory using the Ubuntu apt install: | `/var/lib/tomcat#/` |
| Home directory using the Linux Jamf Pro installer: | `/usr/local/jss/tomcat/` |
| Catalina log on Ubuntu in the Tomcat home directory: | `.../logs/catalina.currentdate.log` |
| `server.xml` on Ubuntu in the Tomcat home directory: | `.../conf/server.xml` |
| Home directory using the Windows Tomcat installer: | `C:\Program Files\Apache Software Foundation\Tomcat #.#\` |
| Home directory using the Windows Jamf Pro installer: | `C:\Program Files\JSS\Tomcat\` |
| Catalina log on Windows in the Tomcat home directory: | `...\Logs\catalina.currentdate.log` |
| `server.xml` on Windows in the Tomcat home directory: | `...\conf\server.xml` |

**The Jamf Pro `DataBase.xml` file stores information required by the web application for communicating with the database**

| | |
|---|---|
| On Ubuntu in the Tomcat home directory: | `.../webapps/contextname/WEB-INF/xml/DataBase.xml` |
| On Windows in the Tomcat home directory: | `...\webapps\contextname\WEB-INF\xml\DataBase.xml` |

**Jamf Pro log locations using the Linux installer**

```
/usr/local/jss/logs/JAMFChangeManagement.log
/usr/local/jss/logs/JAMFSoftwareServer.log
/usr/local/jss/logs/JSSAccess.log
```

**Jamf Pro log locations using the Windows Installer**

```
C:\Program Files\JSS\Logs\JAMFChangeManagement.log
C:\Program Files\JSS\Logs\JAMFSoftwareServer.log
C:\Program Files\JSS\Logs\JSSAccess.log
```

**The Jamf Pro `log4.properties` file defines log locations when Tomcat has been installed manually**

| | |
|---|---|
| On Ubuntu in the Tomcat home directory: | `.../webapps/contextname/WEB-INF/classes/log4j.properties` |
| On Windows in the Tomcat home directory: | `...\webapps\contextname\WEB-INF\classes\log4j.properties` |

# Commands

## Java

| | |
|---|---|
| Determine Windows Tomcat version | `java -cp "...\lib\catalina.jar" org.apache.catalina.util.ServerInfo` |
| Determine Java version | `java -version` |
| View keytool commands and subcommands | `keytool -command -help [subcommand]` |
| Delete a specific alias inside the keystore file | `sudo keytool -delete -alias aliasname -keystore keystore.jks` |
| Create a public / private key pair in a keystore file | `sudo keytool -genkeypair -alias aliasname -keystore keystore.jks` |
| Import a certificate file into the keystore file | `sudo keytool -import -trustcacerts -alias aliasname -file file.pem -keystore keystore.jks` |
| View the contents of a keystore file | `sudo keytool -list -v -keystore keystore.jks` |

## nano

| | |
|---|---|
| Display cursor location (line count) | `sudo nano -c /filepath` |
| Create backup copy of file selected for editing | `sudo nano -B /filepath` |
| Place a backup copy in a specific directory | `sudo nano -C /filepath` |
| Save find and replace history in a log for reuse | `sudo nano -H /filepath` |
| **Editor commands:** Find and replace | `CONTROL + \ (^\)` |
| Display line count | `CONTROL + C (^C)` |
| Remove (cut) a single line | `CONTROL + K (^K)` |
| Search for a text string | `CONTROL + W (^W)` |

## MySQL Command Line Interface

| | |
|---|---|
| Create a database | `CREATE DATABASE dbname;` |
| Delete an account | `DROP USER 'username'@'hostname';` |
| Create account / Grant access for an account to a database | `GRANT ALL ON dbname.* TO 'username'@'hostname' IDENTIFIED BY 'password';` |
| Revoke access to an account on a specific database | `REVOKE ALL PRIVILEGES ON dbname.* from 'username'@'hostname';` |
| List all accounts and hashed passwords | `SELECT USER, HOST, AUTHENTICATION_STRING FROM mysql.user;` |
| Determine MySQL version | `SELECT VERSION();` |
| List all databases | `SHOW DATABASES;` |
| View server process threads / connections | `SHOW PROCESSLIST;` or `SHOW FULL PROCESSLIST\G` |

| | |
|---|---|
| View all tables in a database | `SHOW TABLES;` |
| View database grants for an account | `SHOW GRANTS FOR 'username'@'hostname';` |
| View the current state of remote MySQL server access | `SHOW VARIABLES LIKE 'bind_address';` |
| View the maximum MySQL server connections value | `SHOW VARIABLES LIKE 'max_connections';` |
| Access a database | `USE dbname;` |

## Ubuntu

| | |
|---|---|
| Install the default Java Developer Kit | `sudo apt install default-jdk` |
| Install a load balancer | `sudo apt install haproxy` |
| Install distributed caching | `sudo apt install memcached` |
| Install MySQL | `sudo apt install mysql-server` |
| Install openSSH | `sudo apt install openssh-server` |
| Install Tomcat | `sudo apt install tomcat#` |
| Install Tomcat Manager | `sudo apt install tomcat#-admin` |
| Install unzip | `sudo apt install unzip` |
| Update the downloadable package repository index | `sudo apt update` |
| Change file ownership | `sudo chown username:groupname /filepath` |
| Create a backup copy of a file | `sudo cp /filepath{,.bak}` |
| Find the Tomcat group name | `sudo cat /etc/group \| grep tomcat` |
| Find the Tomcat user name | `sudo cat /etc/passwd \| grep tomcat` |
| Determine the Jamf Pro version | `cat .../WEB-INF/xml/version.xml \| grep jamfWebApplication -A2` |
| Set the date and time | `sudo date --set="YYYY-MM-DD HH:MM:SS.000"` |
| Set the time zone | `sudo dpkg-reconfigure tzdata` |
| Create a directory | `sudo mkdir -p /filepath` |
| Determine the MySQL version | `sudo mysqladmin -u root -p version` |
| Login to the MySQL Command Line Interface as the root user | `mysql -u root -p` |
| Test MySQL account database access | `mysql -u user -h host -p` |
| Restore a backup copy of a MySQL database | `sudo mysql -u root -p -v dbname < dbname.sql` |
| Create a backup copy of a MySQL database | `sudo mysqldump -u root -p --dump-date --lock-tables dbname > dbname.sql` |

| | |
|---|---|
| Listen for a service running on a remote host | `nc -z IPADDRESS PORT` |
| Delete a file or directory | `sudo rm -rf /filepath` |
| Securely copy a file to a remote host | `scp filepath username@hostname:/dirname/` |
| Securely copy a file from a remote host | `scp username@hostname:filepath /dirname/` |
| List all service names and statuses | `sudo service --status-all` |
| Restart service | `sudo service servicename restart` |
| Start service | `sudo service servicename start` |
| Check service status | `sudo service servicename status` |
| Stop service | `sudo service servicename stop` |
| Determine Tomcat version (Jamf Pro installer) | `sudo sh /usr/local/jss/tomcat/bin/version.sh` |
| Determine Tomcat version (manual install) | `sudo sh /usr/share/tomcat8/bin/version.sh` |
| Remote secure shell login | `ssh username@hostname` |
| Create a blank file | `sudo touch /filepath` |

## Windows

| | |
|---|---|
| Determine Jamf Pro version (PS) | `GC "...\WEB-INF\xml\version.xml" \| Select-String -Pattern "jamfWebApplication" -Context 0,2` |
| Restore a backup copy of a MySQL database | `mysql -u root -p -v dbname < dbname.sql` |
| Determine MySQL version | `mysqladmin -u root -p version` |
| Create a backup copy of a MySQL database | `mysqldump -u root -p --dump-date --lock-tables dbname > dbname.sql` |
| Start service | `net start servicename` |
| Stop service | `net stop servicename` |
| Add a Windows Firewall rule | `netsh advfirewall firewall add rule name=name dir=in action=allow protocol=tcp localport=##` |
| List all Windows Firewall rules | `netsh advfirewall firewall show rule name=all` |
| Delete a Windows Firewall rule | `netsh advfirewall firewall delete rule name=name protocol=tcp localport=##` |
| Allocate Tomcat memory | `"C:\...\bin\Tomcat#.exe" //US//tomcat# --JvmMs=##MB --JvmMx=##MB` |
| Install the Tomcat Properties application on the Task Bar | `"C:\...\bin\Tomcat#w.exe" //MS//Tomcat#` |

# Installation and Configuration

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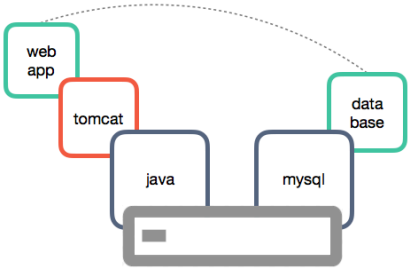
# Jamf Pro Installer




The Jamf Pro Installer ***does***:

- Install the latest Jamf-supported version of Apache Tomcat
- Install a Tomcat home directory inside the JSS home directory

**macOS:** `/Library/JSS/Tomcat/`
**Linux:** `/usr/local/jss/tomcat/`
**Windows:** `C:\Program Files\JSS\Tomcat\`

- Deploy a web application with logging correctly enabled for the host operating system
- Create a self-signed SSL certificate keystore for testing the installation

The Jamf Pro Installer ***does not***:

- Install Java or the Java Cryptography Extensions
- Install MySQL
- Create a MySQL database
- Create a MySQL account for the web application

In general, use the Jamf Pro Installer for the following scenarios:

- One server hosting the web application and the database *
- One server hosting the web application and a different server hosting the database **
- Multiple servers each hosting a web application all connecting to one database ***

In general, ***do not*** use the Jamf Pro Installer for the following scenarios:

- The intended host server is running an unsupported operating system
- A different version of Tomcat is required than the one installed by the Jamf Pro installer
- There is potential conflict with already installed web services on the intended host
- The intended Tomcat server will host multiple web applications ****

basic configuration *

basic configuration + **

clustered environment ***

multi-context environment ****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# Lesson 1: Windows MySQL Installation and Configuration

## Objective

Install MySQL on Windows. Create a database for the web application. Grant access from the web application to the database by creating a MySQL account.

## Discussion

- MySQL Installer for Windows
- MySQL Command Syntax
- MySQL Account Names: `'username'@'hostname'`

## Tasks

COMPLETE THE FOLLOWING STEPS ON WINDOWS VM 3

1. Install MySQL.

   a. Open the MySQL Installer and accept the license terms.
   b. Select the "Server only" option on the "Choosing a Setup Type" screen then click "Next".
   c. Follow the installer prompts.
   d. On the "Type and Networking" screen select "Server Machine" from the "Config Type" menu.
   e. Enter a password for the MySQL root user.
   f. **OPTIONAL:** Change the "Windows Service Name" to `mysql`.
   g. When the all installer steps are complete click "Finish".

2. Create a database.

   a. Open the MySQL CLI client and log in as the root user created during the installation.
   b. Execute the following command:

```
CREATE DATABASE jamfdb;
```

3. Grant access to the `jamfdb` database:

```
GRANT ALL ON jamfdb.* TO jamf@localhost IDENTIFIED BY 'Jamf1234';
```

## End State

- MySQL is installed on Windows
- A database for the web application is created
- A MySQL account for the web application is granted access to the database

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# Lesson 2: Windows Tomcat Installation and Memory Configuration

## Objective

Install Java and Tomcat on Windows. Allocate memory for Tomcat.

## Discussion

- Tomcat dependency on Java
- Tomcat as a Windows service
- Tomcat default web application
- Memory allocation
- Windows command line: `cmd`

## Tasks

COMPLETE THE FOLLOWING STEPS ON WINDOWS VM 3

1. Install Java.

   a. Open the Java SE JDK Installer.
   b. Use the Installation Wizard to install Java with the default options.

2. Install the Java Cryptography Extensions (JCE).

   a. Locate and open the `UnlimitedJCEPolicyJDK8` directory in the Student Resources directory.
   b. Select the `local_policy` and `US_export_policy` files.
   c. Replace the identically named files in the directory below with the selected files:

```
C:\Program Files\Java\jre1.8.#_###\lib\security
```

3. Install Tomcat.

   a. Open the Tomcat Installer.
   b. Use the Installation Wizard to install Tomcat.
   c. On the Choose Components pane click Add (+) next to Tomcat and select "Service Startup".
   d. On the Choose Components pane leave the "Manager" checkbox selected.
   e. On the Configuration pane create a username and password for Tomcat Manager.

4. **OPTIONAL:** Navigate to `http://192.168.56.103:8080/` using a Windows-compatible 3rd party web browser installed from the Student Resources directory to view the Tomcat default web application.


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5. Set memory for Tomcat on the Java tab of the Tomcat Properties application located on the Notification Center Task Bar at the bottom right of the screen using the following values:

Initial memory pool: 512 MB
Maximum memory pool: 512 MB

or execute the following command on the Windows command line:

```
"C:\Program Files\Apache Software Foundation\Tomcat 8.0\bin\Tomcat8.exe" //US//tomcat8 --
JvmMs=512 --JvmMx=512
```

The Java options in the table below can be set in the Options window of the Tomcat Properties application or on the command line.

| Attribute | Description |
|---|---|
| `-XX:MaxPermSize` | Maximum Permanent Generation memory **(Java 7 or prior)** |
| `-XX:MaxMetaspaceSize` | Maximum MetaSpace memory **(Java 8 or later)** |

6. Restart Tomcat on the General tab of the Tomcat Properties application located on the Notification Center Task Bar at the bottom right of the screen. If the application is not on the Task Bar it can be added by executing the following on the Windows command line:

```
"C:\Program Files\Apache Software Foundation\Tomcat 8.0\bin\tomcat8w.exe" //MS//Tomcat8
```

Tomcat can also be restarted by entering the following commands on the Windows command line using the name given to the Tomcat service:

```
net stop servicename
net start servicename
```

## End State

- Java is installed on Windows
- Tomcat is installed on Windows
- Additional memory is allocated to Tomcat on Windows

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# Lesson 3: Windows Manual Web Application Installation

## Objective

Manually install the Jamf Pro web application. Configure access to the web application through the Windows firewall.

## Discussion

- `ROOT.war`
- Windows Firewall
- Database Connection Error page
- Jamf Pro Setup Assistant

## Tasks

COMPLETE THE FOLLOWING STEPS ON WINDOWS VM 3

1. Copy the `ROOT.war` file from the Student Resources directory to the Windows Desktop.

2. Delete the directory named `ROOT` in the Tomcat webapps directory:

```
C:\Program Files\Apache Software Foundation\Tomcat 8.0\webapps\ROOT
```

3. Install the web application by placing the `ROOT.war` file in the Tomcat webapps directory:

```
C:\Program Files\Apache Software Foundation\Tomcat 8.0\webapps
```

It can take several minutes for the `ROOT.war` to explode. If the `ROOT.war` fails to explode execute the following commands on the Windows command line:

```
mkdir "C:\Program Files\Apache Software Foundation\Tomcat 8.0\webapps\ROOT"

cd "C:\Program Files\Apache Software Foundation\Tomcat 8.0\webapps\ROOT"

"C:\Program Files\Java\jdk1.8.0_121\bin\jar.exe" xvf "C:\Users\Administrator\Desktop\ROOT.
war"
```

4. Make a firewall exception for port `8080` to enable remote access to Tomcat by executing the following command on the Windows command line:

```
netsh advfirewall firewall add rule name=tomcat8080 dir=in action=allow protocol=tcp
localport=8080
```

5. Restart Tomcat on the General tab of the Tomcat Properties application or using any preferred method.

6. Browse to `http://192.168.56.103:8080/` to verify the installation.


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7. Click **Edit Connection** on the Database Connection Error page.

   a. Modify information to match the account and database created in the MySQL CLI in Lesson 1.

8. Complete the Jamf Pro Setup Assistant.

   a. Click **Agree** on the License Agreement page.
   b. Enter an Organization Name.
   **NOTE:** The Organization Name permanently populates the Jamf Pro Certificate Authority.
   c. Enter the Activation Code provided by the instructor located in the Student Resources directory.
   d. Create a Jamf Pro User (carefully note the credentials entered using any preferred method.)
   e. Enter `https://vm3.jamf.training:8443/` as the Jamf Pro URL for upcoming lessons.

## End State

- The Jamf Pro `ROOT.war` file has been installed in the Tomcat webapps directory
- The Windows firewall has been configured for remote access on port `8080`
- The configured Jamf Pro server is available via its web interface

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# Lesson 4: Windows Tomcat Port Configuration

## Objective

Modify the default `http` port configuration for Tomcat on Windows.

## Discussion

- Use cases for changing ports
- `server.xml`
- Windows Firewall with Advanced Security GUI

## Tasks

COMPLETE THE FOLLOWING STEPS ON WINDOWS VM 3

1. Install Notepad++ which can be found in the Student Resources folder.

2. Locate the `server.xml` file in the following location:

```
C:\Program Files\Apache Software Foundation\Tomcat 8.0\conf\server.xml
```

3. Make a backup copy of the `server.xml` on the Windows Desktop or any accessible location.

4. Modify the `server.xml` file by changing the `http` port value.

   a. Open the `server.xml` file with Notepad ++. (Avoid using NotePad or WordPad.)
   b. Change the value from `8080` to `9006` in the active `Connector` port element:

```
<Connector port="9006"
           protocol="HTTP/1.1"
           connectionTimeout="20000"
           redirectPort="8443" />
```

5. Restart Tomcat.

6. Make a firewall exception for port `9006` to enable remote access to Tomcat using the Windows Firewall GUI or by executing the following command on the Windows command line:

```
netsh advfirewall firewall add rule name=tomcat9006 dir=in action=allow protocol=tcp
localport=9006
```

7. Browse to `http://192.168.56.103:9006/` to verify the configuration.

8. Change the port value from `"9006"` back to `"8080"`.


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9. Restart Tomcat.

10. Check all Windows firewall rules with the following command to determine which ports are open:

```
netsh advfirewall firewall show rule name=all
```

11. Disable the exception for port `9006` with the following command:

```
netsh advfirewall firewall delete rule name=tomcat9006 protocol=tcp localport=9006
```

## End State

- The web application is made available on port `9006` and reverted back to port `8080`
- The Windows firewall is modified to match the ports intended for Tomcat traffic

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# Lesson 5: Ubuntu MySQL Installation and Configuration

## Objective

Install MySQL on Ubuntu. Create a database for the web application. Grant access from the web application to the database by creating a MySQL account.

## Discussion

- Unix shell and command prompt
- macOS Terminal.app
- Ubuntu commands: `apt`, `ssh`, `sudo`
- Preparing a clustered environment

## Tasks

COMPLETE THE FOLLOWING STEPS ON UBUNTU VM 1

1. Execute the following commands in the Ubuntu VM 1 VMware Fusion.app window:

```
sudo apt update
sudo apt install openssh-server
```

2. Open a macOS Terminal.app window and execute the following command - when prompted enter `yes` then enter the password for the `serveradmin` user.

```
ssh serveradmin@192.168.56.101
```

**NOTE:** from this point forward using an `ssh` session to access Ubuntu is preferred. The `ssh` service must be installed on the virtual machine to enable remote access.

3. Install MySQL by executing the following command - when prompted during the installation create a password for the MySQL root user.

```
sudo apt install mysql-server
```

4. Log in to the MySQL CLI by executing the following command - when prompted use the password for the MySQL root user created during the installation above:

```
mysql –u root -p
```


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5. Create a database by executing the following command:

```
CREATE DATABASE jamfcluster;
```

6. Grant access to the `jamfcluster` database:

```
GRANT ALL ON jamfcluster.* TO jamf@localhost IDENTIFIED BY 'jamf101';
```

7. Return to the Ubuntu command line by typing `exit` at the MySQL command line prompt.

## End State

- MySQL is installed on Ubuntu
- A database for the web application is created
- A MySQL account for the web application is granted access to the database

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# Lesson 6: Ubuntu Tomcat Installation and Memory Configuration

## Objective

Install Java and Tomcat on Ubuntu. Allocate memory for Tomcat.

## Discussion

- Tomcat `CATALINA` process
- Tomcat environment variable configuration
- Ubuntu commands: `chmod`, `grep`, `nano`, `ps`, `service`

## Tasks

COMPLETE THE FOLLOWING STEPS ON UBUNTU VM 1

1. Install Java and Tomcat.

   a. Execute the `sudo apt update` command if it has not already been run.
   b. Execute the following command to complete the installation:

```
sudo apt install tomcat8
```

2. Allocate memory to Tomcat by using `nano` to create an environment variable script:

```
sudo nano /usr/share/tomcat8/bin/setenv.sh
```

3. Enter the following text into the `nano` window exactly as formatted:

```
#!/bin/bash
export CATALINA_OPTS="-Xms512M -Xmx512M -Djava.awt.headless=true -server"
```

Option definitions used in the script above:

| Attribute | Description |
|---|---|
| `-Xms` | Minimum heap size for memory used by Java |
| `-Xmx` | Maximum heap size for memory used by Java |
| `-Djava.awt.headless` | Allows Java to run on systems with no display, keyboard, or mouse |
| `-server` | Instructs the launcher to use the Java HotSpot Server VM |


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4. Make the `setenv.sh` script executable:

```
sudo chmod 755 /usr/share/tomcat8/bin/setenv.sh
```

5. Use the following command to view the current Java memory settings:

```
ps aux | grep java
```

6. Restart Tomcat to read in the memory allocation changes from the `setenv.sh` script:

```
sudo service tomcat8 restart
```

7. When Tomcat is active execute the `ps aux | grep java` command again to view the modified `CATALINA` memory settings.

## End State

- Java is installed on Ubuntu
- Tomcat is installed on Ubuntu
- Additional memory is allocated to Tomcat on Ubuntu

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# Lesson 7: Ubuntu Manual Web Application Installation

## Objective

Manually install the Jamf Pro web application. Configure User-Initiated Enrollment.

## Discussion

• Ubuntu commands: `cd`, `chown`, `cp`, `ls -lsa`, `mkdir`, `rm`, `scp`, `zip` / `unzip`

## Tasks

1. Copy the `ROOT.war` file from the Mac to Ubuntu VM 1:

   a. Locate the `ROOT.war` file in the Student Resources directory and copy it to the Desktop.
   b. Open Terminal.app on the Mac if it is not already open and execute the following command:

```
scp ~/Desktop/ROOT.war serveradmin@192.168.56.101:/tmp/
```

COMPLETE THE FOLLOWING STEPS ON UBUNTU VM 1

2. Delete the directory named `ROOT` in the Tomcat `webapps` directory:

```
cd /var/lib/tomcat8/webapps/
sudo rm -rf ROOT/
```

3. Install the web application by placing the `ROOT.war` file in the Tomcat `webapps` directory:

```
sudo cp /tmp/ROOT.war /var/lib/tomcat8/webapps/
```

4. Determine if the `ROOT.war` file has exploded:

```
ls -lsa /var/lib/tomcat8/webapps/
```

It can take several minutes for the `ROOT.war` to explode. If the `ROOT.war` fails to explode execute the following (4) commands:

```
sudo apt install zip
sudo mkdir /var/lib/tomcat8/webapps/ROOT/
sudo unzip /tmp/ROOT.war -d /var/lib/tomcat8/webapps/ROOT/
sudo chown tomcat8:tomcat8 /var/lib/tomcat8/webapps/ROOT/
```


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5. Restart Tomcat by executing the following command:

```
sudo service tomcat8 restart
```

6. Browse to `http://192.168.56.101:8080/` to verify the installation.

7. Click **Edit Connection** on the Database Connection Error page.

   a. Modify information to match the account and database created in the MySQL CLI in Lesson 5.

8. Complete the Jamf Pro Setup Assistant using `https://pro.jamf.training:8443` as the Jamf Pro URL for upcoming lessons.

9. Enable User-Initiated Enrollment.

   a. Click **Settings**.
   b. Go to **Global Management** > **User-Initiated Enrollment**.
   c. Click on the **Platforms** tab.
   d. Click **macOS** then click **Edit**.
   e. Check **Enable user-initiated enrollment for computers** and complete all required fields.

## End State

- The Jamf Pro `ROOT.war` file has been installed in the Tomcat webapps directory
- The configured Jamf Pro server is available via its web interface
- User-Initiated Enrollment is enabled

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# Lesson 8: Ubuntu Tomcat Port Configuration

## Objective

Modify the default `http` port configuration for Tomcat on Ubuntu.

## Discussion

- Unix filesystem management
- Ubuntu commands: `cp`, `mv`

## Tasks

COMPLETE THE FOLLOWING STEPS ON UBUNTU VM 1

1. Create a backup copy of the `server.xml` file using a command like the following:

```
sudo cp /var/lib/tomcat8/conf/server.xml{,.bak}
```

**NOTE:** File extensions can literally be any string that helps to organize backup copies, e.g., `.backup` `.copy` `.good` `.old` `.kg` `.bad` etc.

2. Use the `mv` command to rename a file, e.g.,

```
cd /var/lib/tomcat8/conf/
sudo mv server.xml.bad server.xml.good
```

3. Open the `server.xml` file in `nano` or in a compatible text editor and modify it by changing the `http` port:

```
sudo nano -Bc /var/lib/tomcat8/conf/server.xml
```

**NOTE:** The `nano -B` option creates a "backup" copy of the file.

4. Change the value from `8080` to `9006` in the active `Connector` port element:

```
<Connector port="9006"
           protocol="HTTP/1.1"
           connectionTimeout="20000"
           redirectPort="8443" />
```

5. Restart Tomcat.

6. Browse to `http://192.168.56.101:9006/` to verify the configuration.

## End State

- The web application is made available on `9006`


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# Lesson 9: Web Application Logging Configuration

## Objective

Configure web application logging on Windows and on Ubuntu.

## Discussion

- Tomcat log files
- Jamf log files
- `log4j.properties`

## Tasks

COMPLETE THE FOLLOWING STEPS ON UBUNTU VM 1

1. Open the `log4j.properties` file using `nano` or a compatible text editor:

```
sudo nano -Bc /var/lib/tomcat8/webapps/ROOT/WEB-INF/classes/log4j.properties
```

2. Locate each of the (3) instances of the following path:

```
/Library/JSS/Logs/
```

3. Use the find and replace (`CONTROL + \`) features in `nano` or any other method to modify the target file path for each of the logs so they read exactly as follows:

```
/var/lib/tomcat8/logs/JAMFChangeManagement.log

/var/lib/tomcat8/logs/JAMFSoftwareServer.log

/var/lib/tomcat8/logs/JSSAccess.log
```

4. Save the file and exit `nano` (CONTROL + x)

5. Restart Tomcat.

COMPLETE THE FOLLOWING STEPS ON WINDOWS VM 3

1. Navigate to the `log4j.properties` file inside of the web application directory:

```
C:\Program Files\Apache Software Foundation\Tomcat 8.0\webapps\ROOT\WEB-INF\classes\log4j.
properties
```

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2. Make a backup copy of the `log4j.properties` file.

3. Open the `log4j.properties` with Notepad++ and locate each of the (3) instances of the following path:

```
/Library/JSS/Logs/
```

4. Use the Find... and Replace... features of Notepad++ or any other method to modify the target file path for each of the logs using one of the following two formats:

```
/Program Files/Apache Software Foundation/Tomcat 8.0/logs/JAMFChangeManagement.log

/Program Files/Apache Software Foundation/Tomcat 8.0/logs/JAMFSoftwareServer.log

/Program Files/Apache Software Foundation/Tomcat 8.0/logs/JSSAccess.log
```

```
\\Program Files\\Apache Software Foundation\\Tomcat 8.0\\logs\\JAMFChangeManagement.log

\\Program Files\\Apache Software Foundation\\Tomcat 8.0\\logs\\JAMFSoftwareServer.log

\\Program Files\\Apache Software Foundation\\Tomcat 8.0\\logs\\JSSAccess.log
```

5. Save the file.

6. Restart Tomcat.

## End State

• Web application log files are written to the preferred location


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# Multi-context

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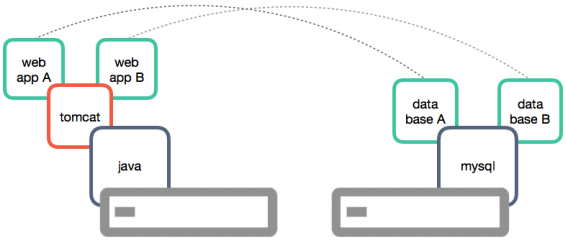
# Multi-context Environments




Tomcat's documentation refers to web applications as "contexts". More than one web application, or, context, in the Tomcat `webapps` directory implies a "multi-context" environment.

By default the Tomcat `webapps` directory can contain multiple contexts like the Tomcat "Docs" web application, the Tomcat "Manager" web application, and, the Tomcat default "ROOT" web application.

Multiple instances of the Jamf Pro web application can also be hosted on a single instance of Tomcat server. Jamf Pro instances configured in this way do not communicate with each other. Each has its own database (unless they are configured as a node in a cluster.)

Multi-context use cases:

• A managed service provider would like to host as many separate Jamf Pro instances as possible on a single Tomcat server.

• An organization needs to create a lab environment for training Jamf administrators.

• An organization wishes to create a test environment by installing additional Jamf Pro instances on the Tomcat server that hosts their production server.

• A complex organization with many independent entities requires each to have its own separate Jamf Pro instance.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# Lesson 10: Multi-context Database Configuration

## Objective

Create unique databases for separate web applications (contexts) on Windows and on Ubuntu. Grant access from the web applications to the databases by creating unique MySQL accounts.

## Discussion

- MySQL command syntax
- Data placeholders
- Unique passwords for MySQL accounts

## Tasks

COMPLETE THE FOLLOWING STEPS ON WINDOWS VM 3

1. Create an additional database.

   a. Open the MySQL CLI client and log in as the root user.
   b. Check the list of databases that already exist by executing the following command:

```
SHOW DATABASES;
```

   c. Execute the following command using a unique database name:

```
CREATE DATABASE databasename;
```

2. Grant access to the database created above:

```
GRANT ALL ON databasename.* to 'username'@'localhost' IDENTIFIED BY 'password';
```

COMPLETE THE FOLLOWING STEPS ON UBUNTU VM 1

1. Create an additional database.

   a. Open the MySQL CLI and log in as the root user created during installation:

```
mysql -u root -p
```

   b. Execute the following command using `testdb` as the database name:

```
CREATE DATABASE testdb;
```

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2. Grant access to the `testdb` database:

```
GRANT ALL ON testdb.* to 'test'@'localhost' IDENTIFIED BY 'test101';
```

## End State

- An additional database for a separate web application is configured on Windows
- An additional database for a separate web application is configured on Ubuntu
- MySQL accounts for the web applications are granted access to the databases


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# Lesson 11: Multi-context Web Application Installation

## Objective

Manually install additional web applications (contexts) on Windows and Ubuntu.

## Discussion

- URL syntax
- `DataBase.xml` / Database Connection Error page
- Multi-context logging configuration

## Tasks

COMPLETE THE FOLLOWING STEPS ON WINDOWS VM 3

1. Prepare the `ROOT.war` file to be installed as a unique Tomcat context.

   a. If needed, copy the `ROOT.war` file to the Windows Desktop.
   b. Give the file a unique name formatted as follows:

```
context.war
```

2. Install the web application by placing the renamed `.war` file in the Tomcat webapps directory:

```
C:\Program Files\Apache Software Foundation\Tomcat 8.0\webapps
```

3. Locate the `DataBase.xml` file in the following location:

```
C:\Program Files\Apache Software Foundation\Tomcat 8.0\webapps\context\WEB-INF\xml\DataBase.
xml
```

   a. Make a backup copy of the `DataBase.xml` file.
   b. Open the `DataBase.xml` file with Notepad ++. (Avoid using NotePad or WordPad.)
   c. Edit the file to match the MySQL account created on Windows VM 3 in Lesson 10:

```
<DataBaseName>databasename</DataBaseName>
<DataBaseUser>username</DataBaseUser>
<DataBasePassword>password</DataBasePassword>
```

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4. Navigate to the `log4j.properties` file inside of the web application directory:

```
C:\Program Files\Apache Software Foundation\Tomcat 8.0\webapps\context\WEB-
INF\classes\log4j.properties
```

5. Make a backup copy of the `log4j.properties` file

6. Open the `log4j.properties` with Notepad++ and locate each of the (3) instances of the following path:

```
/Library/JSS/Logs/
```

7. Use the Find... and Replace... features of Notepad++ or any other method to modify the target file path for each of the log files. The modified paths should include a separate folder with the same name as the context name used in step 1 to prevent overwriting, e.g.,

```
/Program Files/Apache Software Foundation/Tomcat 8.0/logs/context/JAMFChangeManagement.log

/Program Files/Apache Software Foundation/Tomcat 8.0/logs/context/JAMFSoftwareServer.log

/Program Files/Apache Software Foundation/Tomcat 8.0/logs/context/JSSAccess.log
```

Either the format above or the one below can be used for Windows file paths:

```
\\Program Files\\Apache Software Foundation\\Tomcat 8.0
\\logs\\context\\JAMFChangeManagement.log

\\Program Files\\Apache Software Foundation\\Tomcat 8.0\\logs\\context\\JAMFSoftwareServer.
log

\\Program Files\\Apache Software Foundation\\Tomcat 8.0\\logs\\context\\JSSAccess.log
```

8. Restart Tomcat using the Tomcat Properties application located on the Notification Center Task Bar at the bottom right of the screen or any preferred method.

9. Browse to `http://192.168.56.103:8080/context` to verify the installation.

10. Use `https://vm3.jamf.training:8443/context` as the Jamf Pro URL to complete the Jamf Pro Setup Assistant.

**COMPLETE THE FOLLOWING STEPS ON UBUNTU VM 1**

1. If needed, copy the `ROOT.war` file from the Mac to Ubuntu VM 1 with a command like:

```
scp ~/Desktop/ROOT.war serveradmin@192.168.56.101:/tmp/
```

2. Use the file name `testing` to copy the `ROOT.war` file from `/tmp/` into the Tomcat webapps directory:

```
sudo cp /tmp/ROOT.war /var/lib/tomcat8/webapps/testing.war
```


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3. Open the `DataBase.xml` file with a text editor, e.g.,

```
sudo nano -Bc /var/lib/tomcat8/webapps/testing/WEB-INF/xml/DataBase.xml
```

a. Edit the file to match the MySQL account created on Ubuntu VM 1 in Lesson 10:

```
<DataBaseName>testdb</DataBaseName>
<DataBaseUser>test</DataBaseUser>
<DataBasePassword>test101</DataBasePassword>
```

The Database Connection Error page can be used in step 8 below to edit MySQL account information instead - either are sufficient for initiating authentication to the database.

4. Open the `log4j.properties` file with a text editor, e.g.,

```
sudo nano -Bc /var/lib/tomcat8/webapps/testing/WEB-INF/classes/log4j.properties
```

5. Locate each of the (3) instances of the following path:

```
/Library/JSS/Logs/
```

6. Use the find and replace (`CONTROL + \`) features in `nano` or any other method to modify the target file path for each of the logs so they read exactly as follows:

```
/var/lib/tomcat8/logs/testing/JAMFChangeManagement.log

/var/lib/tomcat8/logs/testing/JAMFSoftwareServer.log

/var/lib/tomcat8/logs/testing/JSSAccess.log
```

The modified paths include a separate folder named `testing` which matches the context name used in step 2 to prevent overwriting.

7. Restart Tomcat by executing the following command:

```
sudo service tomcat8 restart
```

8. Browse to `http://192.168.56.101:9006/testing` to verify the installation.

## End State

- Tomcat on Windows hosts multiple Jamf Pro web applications / contexts
- Tomcat on Ubuntu hosts multiple Jamf Pro web applications / contexts
- Multi-context web application log files are written to separate locations

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# Lesson 12: Tomcat Manager

## Objective

Enable Tomcat Manager for local access on Windows.

## Discussion

- Capabilities
- Security risks

## Tasks

COMPLETE THE FOLLOWING STEPS ON WINDOWS VM 4

1. Install Java.

   a. Open the Java SE JDK Installer.
   b. Use the Installation Wizard to install Java with the default options.

2. Install Tomcat version 8.5:

   a. Open the Tomcat 8.5 Installer.
   b. Use the Installation Wizard to install Tomcat.
   c. On the Choose Components pane click Add (+) next to Tomcat and select "Service Startup".
   d. On the Choose Components pane leave the "Manager" checkbox selected.
   e. On the Configuration pane create a username and password for Tomcat Manager.

3. Navigate to `http://192.168.56.104:8080/` using a Windows-compatible 3rd party web browser installed from the Student Resources directory to view the Tomcat default web application.

4. Navigate to the Tomcat webapps directory and take note of the installed web applications / contexts.

```
C:\Program Files\Apache Software Foundation\Tomcat 8.5\webapps\
```

5. Locate the `server.xml` file in the Tomcat home directory and open it using Notepad++ or any compatible text editor:

```
C:\Program Files\Apache Software Foundation\Tomcat 8.5\conf\server.xml
```

6. Locate the active `connector` element.

```
<Connector port="8080"
           protocol="HTTP/1.1"
           connectionTimeout="20000"
           redirectPort="8443" />
```


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a. Add the `address` attribute using `192.168.56.104` and `127.0.0.1` as the values.
b. Add the `useIPVHosts` attribute using `true` as the value.

```
<Connector port="8080"
           protocol="HTTP/1.1"
           connectionTimeout="20000"
           redirectPort="8443"
           address="192.168.56.104,127.0.0.1"
           useIPVHosts="true" />
```

c. Save and close the file.

7. Locate the `context.xml` file:

```
C:\Program Files\Apache Software Foundation\Tomcat 8.5\webapps\manager\META-INF\context
```

8. Modify the file to include the IP address of the local computer using the formatting below:

```
<Context antiResourceLocking="false"
         privileged="true" >
  <Valve className="org.apache.catalina.valves.RemoteAddrValve"
         allow="192\.168\.56\.104|127\.\d+\.\d+\.\d+" />
</Context>
```

9. Restart Tomcat.

10. Browse to `http://192.168.56.104:8080/manager` to verify the configuration.

11. Enter the credentials created for Manager during the Tomcat installation.

12. Reset Windows VM 4 in VMware Fusion.app to its Clean snapshot.

## End State

• Tomcat Manager is enabled for local access only and then reset

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# Clustering

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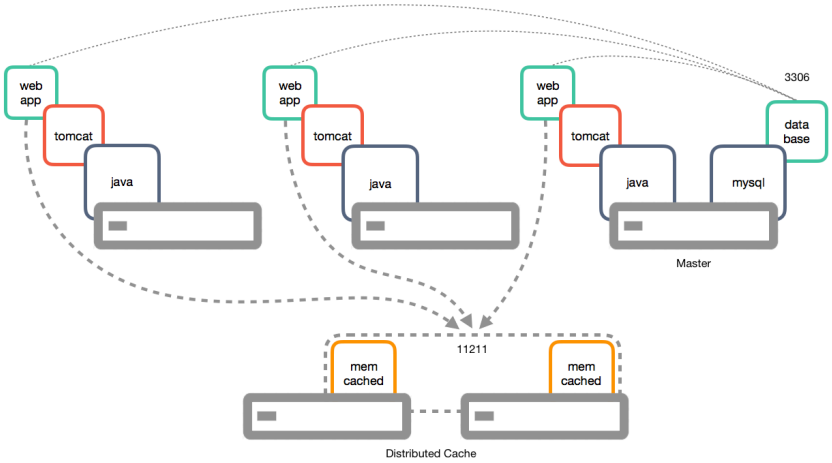
# Clustered Environments




An environment can outgrow the capabilities of a single Jamf Pro web application. If Jamf Pro performance becomes an issue in an environment Clustering can be enabled.

There is no magical device number threshold at which Clustering becomes necessary. Clustering may be needed in an environment with a small number of devices if all Jamf Pro features are heavily used. Clustering may not be needed in an environment with many devices if Jamf Pro is only used for tracking inventory.

Clustering allows traffic to be distributed to multiple Jamf Pro web applications. Clustered web applications authenticate to a single MySQL database. The "cluster" functions as a single Jamf Pro server. This type of architecture is also known as "horizontal scaling." Distributing traffic to multiple web applications effectively reduces the number of check-ins per web application. Raising the average time between check-ins at each Tomcat instance improves performance.

Clustered "nodes" have specialized functions. The "master" node handles the task of synchronizing all data written to the MySQL database and time-based functions like log flushing and message queuing. Client traffic is routed to "child" nodes using a load balancer. Authentication services on child nodes can be limited for security. Specializing child nodes for client traffic frees the master node to also function as a dedicated administrative console for accessing the Jamf Pro GUI.

Clustering requires the installation and configuration of memcached. Memcached is used to create a distributed cache for Jamf Pro data for the purpose of improving MySQL database performance.

Clustering use cases:

• An organization wants to manage devices both inside and outside of their private network.

• A large organization with many devices under management requires horizontal scaling, i.e., multiple web applications behind a load balancer to allow Tomcat to handle increased client traffic.

• A large organization wants to manage devices both inside and outside of their private network and has many devices under management requiring horizontal scaling.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# Lesson 13: Clustering Database Configuration

## Objective

Grant access from web applications in the cluster to the database. Configure MySQL to allow remote access and handle more incoming connections.

## Discussion

- MySQL account names / processes
- MySQL system variables
- `bind-address`
- calculating `max_connections`

## Tasks

COMPLETE THE FOLLOWING STEPS ON UBUNTU VM 1

1. Open the MySQL CLI and log in as the root user:

```
mysql -u root -p
```

2. Grant access to the `jamfcluster` database for all web applications that are not yet configured by entering the following commands:

```
GRANT ALL ON jamfcluster.* TO jamf@192.168.56.102 IDENTIFIED BY 'jamf102';
GRANT ALL ON jamfcluster.* TO jamf@192.168.56.104 IDENTIFIED BY 'jamf104';
```

3. Disable the `bind-address` in the MySQL configuration file to allow database access from remote hosts.

   a. Open the Ubuntu MySQL configuration file in a text editor:

```
sudo nano /etc/mysql/mysql.conf.d/mysqld.cnf
```

   b. Find the text `bind-address` and comment (i.e., disable) the line by adding a `#` character:

```
# bind-address = 127.0.0.1
#
# * Fine Tuning
```

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4. Increase the `max_connections` value in the MySQL configuration file to account for the total number of web applications (4) currently connected to MySQL server on Ubuntu VM 1.

The `max_connections` value is calculated per web application based on the `MaxPoolSize` value (45) found in the `DataBase.xml` file. One additional connection is usually added to ensure a connection to the MySQL CLI , e.g., (4 x 45) + 1 = 181 (See Lesson 31 for more detail on this configuration.)

a. Find the text `max_connections` and remove the `#` at the beginning of the line.
b. Edit the `max_connections` value to match the example below:

```
# * Fine Tuning
#
key_buffer_size = 16M
max_allowed_packet = 16M
thread_stack = 192K
thread_cache_size = 8
# This replaces the startup script and checks MyISAM tables if needed
# the first time they are touched
myisam-recover-options = BACKUP
max_connections = 181
```

**NOTE:** On Windows the MySQL configuration file is located at the following path:

```
C:\ProgramData\MySQL\MySQL Server 5.7\my.ini
```

5. Restart MySQL:

```
sudo service mysql restart
```

6. Open the MySQL CLI and log in as the root user:

```
mysql -u root -p
```

7. Enter the following command to verify the `max_connections` value:

```
SHOW VARIABLES LIKE 'max_connections';
```

## End State

- Additional web applications have been granted access to the `jamfcluster` database
- MySQL system variables are configured


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# Lesson 14: Memcached Installation

## Objective

Enable distributed caching by installing memcached on 2 Ubuntu servers.

## Discussion

- Distributed caching - `Ehcache` / `memcached`
- Requirements in clustered / multi-context environments
- Performance optimization

## Tasks

COMPLETE THE FOLLOWING STEPS ON UBUNTU VM 2 AND UBUNTU VM 5

1. Execute the `sudo apt update` command if it has not already been run.
2. Install `openssh-server` on the Ubuntu VM being configured to enable Terminal.app remote access.
3. Install `memcached` by executing the following command:

```
sudo apt install memcached
```

4. Modify the `memcached` configuration file by opening it in a text editor:

```
sudo nano /etc/memcached.conf
```

a. Increase the amount of memory allocated to the cache:

```
-m 1024
```

b. Change the listener from `127.0.0.1` to the IP address of the `memcached` server being configured.

```
-l 192.168.56.102
```

**NOTE:** eg. use `-l 192.168.56.105` as the listener address on Ubuntu VM 5.

5. Restart `memcached`:

```
sudo service memcached restart
```

6. Ensure that the `memcached` service is running before proceeding to configure the next server:

```
sudo service memcached status
```

## End State

- `memcached` is installed on Ubuntu VM 2 and on Ubuntu VM 5
- Additional memory has been allocated for the cache

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# Lesson 15: Clustering Master Node Configuration

## Objective

Understand and enable Clustering. Configure the master node to use memcached.

## Discussion

- `cache.properties`
- `memcached.properties`
- Clustering GUI
- Sync frequency
- Memcached Nodes on Clustering page
- Selecting the master node on the Clustering page

Clustering Server Information Overview:

| Name | IP Address / FQDN | Function |
|---|---|---|
| Ubuntu VM 1 | `192.168.56.101`<br>`http://192.168.56.101:9006/`<br>`http://vm1.jamf.training:9006/`<br>`http://192.168.56.101:9006/testing` | • database<br>• master node<br>• Jamf Pro GUI<br>• `testing` context |
| Ubuntu VM 2 | `192.168.56.102`<br>`http://192.168.56.102:8080/`<br>`http://vm2.jamf.training:8080/`<br>`https://192.168.56.102:8443/`<br>`https://vm2.jamf.training:8443/` | • child node<br>• self-signed cert<br>• client traffic<br>• backend<br>• memcached |
| Windows VM 4 | `192.168.56.104`<br>`http://192.168.56.104:8080/`<br>`http://vm4.jamf.training:8080/`<br>`https://192.168.56.104:8443/`<br>`https://vm4.jamf.training:8443/` | • child node<br>• self-signed cert<br>• client traffic<br>• backend |
| Ubuntu VM 5 | `192.168.56.105`<br>`https://pro.jamf.training:8443/` | • reverse proxy<br>• load balancer<br>• memcached |


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## Tasks

COMPLETE THE FOLLOWING STEPS ON UBUNTU VM 1

1. Open the `cache.properties` file inside the `ROOT` web application directory with a text editor.

```
sudo nano /var/lib/tomcat8/webapps/ROOT/WEB-INF/classes/dal/cache.properties
```

a. Change the `cache.type` value to `memcached` and delete the `DBAPPLIED_` string, if present, to overwrite previously applied settings in the database.

```
cache.type=memcached
```

b. Save and close the file after making changes.

2. Locate the `memcached.properties` file which can be found in the same directory as the `cache.properties` file.

a. Add the `memcached` endpoints enabled in Lesson 14 by uncommenting and modifying the example lines as follows:

```
# Example:
memcached.endpoints[0]=192.168.56.102:11211
memcached.endpoints[1]=192.168.56.105:11211
```

b. Locate the `memcached.uniqueId` attribute:

```
# Unique id required when sharing memcached endpoints with two different JSS apps
# memcached.uniqueId=JAMF
```

c. Uncomment and then modify the `memcached.uniqueId` value to match the `ROOT` context as follows:

```
memcached.uniqueId=ROOT
```

d. Save and close the file.

3. Locate and open the `cache.properties` file inside the `testing` web application directory with a text editor.

```
sudo nano /var/lib/tomcat8/webapps/testing/WEB-INF/classes/dal/cache.properties
```

a. Modify the `cache.properties` file as in step 1a above.
b. Save and close the file.

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4. Locate the `memcached.properties` file inside the `testing` web application directory:

```
sudo nano /var/lib/tomcat8/webapps/testing/WEB-INF/classes/dal/memcached.properties
```

a. Modify the `memcached.properties` file as in step 2a above.
b. Locate the `memcached.uniqueId` attribute.
c. Uncomment the line, then, modify the value to match the `testing` context as follows:

```
memcached.uniqueId=testing
```

d. Save and close the file.

5. Restart Tomcat.

6. Using the IP address of a server on which memcached is installed enter a command like the following to test communication:

```
telnet 192.168.56.102 11211
```

When connected enter the following command:

```
stats
```

The `stats` command should return something like:

```
STAT pid 106796
STAT uptime 5696
STAT time 1274397461
STAT version 1.4.5
...
END
```

To end the `telnet` session hold down CONTROL and hit the right bracket key, then, type quit at the `telnet>` prompt.

7. Browse to `http://192.168.56.101:9006` then log in to Jamf Pro:

a. In the top-right corner of the page click **Settings**.
b. Click on **Clustering** to verify the **Memcached Nodes** on the bottom of the **Clustering** Page.
c. Click **Edit**.
d. Select Ubuntu VM 1 (192.168.56.101) as the **Master Clustered Node**.
**NOTE:** `localhost` will appear as the **Address** if used in the MySQL grant command.
e. Click **Save**.

## End State

• Jamf Pro Clustering functions are enabled


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# Lesson 16: Clustering Web Application Installation

## Objective

Configure the clustered environment using the Jamf Pro Installer for Linux and the Jamf Pro Installer for Windows.

## Discussion

- Installer downloads
- Installer pros / cons
- Requirements
- Ubuntu commands: `.run` execution, `unzip`

## Tasks

1. Copy the installer file from the Mac to Ubuntu VM 2:

   a. Locate the Jamf Pro Installer for Linux in the Student Resources directory.
   b. Open Terminal.app on the Mac and execute a command like the following:

```
scp /Users/Shared/Resources/Apps/jamf-pro-installer-linux-#.#.#.zip serveradmin@192.
168.56.102:/tmp/
```

2. Open an `ssh` session to Ubuntu VM 2 in Terminal.app:

```
ssh serveradmin@192.168.56.102
```

COMPLETE THE FOLLOWING STEPS ON UBUNTU VM 2

3. Update the Ubuntu package repository index:

```
sudo apt update
```

4. Install Java:

```
sudo apt install default-jdk
```

5. Install the `unzip` binary:

```
sudo apt install unzip
```

6. Expand the Jamf Pro Installer archive file using the `unzip` command:

```
sudo unzip /tmp/jamf-pro-installer-linux-#.#.#.zip -d /tmp/
```

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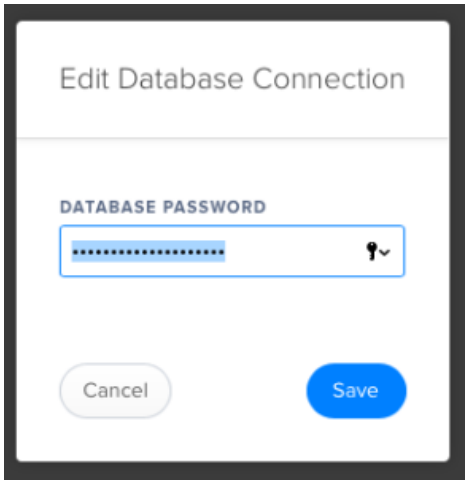
7. Execute the Jamf Pro installer:

```
sudo ./jamfproinstaller.run
```

8. Browse to `http://192.168.56.102:8080` to verify the installation.
9. Set the "Database Hostname" to `192.168.56.101` on the Database Connection Error page to match the information used in Lesson 13 (step 2) when creating the MySQL account for Ubuntu VM 2.

**NOTE:** You may see this popup in the browser - the Database Connection Error page intentionally shows only the password field after the initial authentication as a security feature:




COMPLETE THE FOLLOWING STEPS ON WINDOWS VM 4

1. Install Java and the Java Cryptography Extensions. (See Lesson 2 for detailed instructions.)
2. Locate the Jamf Pro Installer for Windows in the Student Resources directory.
   a. Double-click on the installer to execute.
   b. Follow the installer prompts and select the "Complete" installation.
3. Create a firewall exception for this Jamf Pro instance on port 8080. (See the Reference section or earlier lessons for detailed instructions.)
4. Browse to `http://192.168.56.104:8080` to verify the installation.
5. Set the "Database Hostname" to `192.168.56.101` on the Database Connection Error pageto match the information used in Lesson 13 (step 2) when creating the MySQL account for Windows VM 4.

## End State

- The Jamf Pro Installer for Linux was used to install the web application
- The Jamf Pro Installer for Windows was used to install the web application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# Lesson 17: Clustering Child Node Configuration

## Objective

Enable clustered web applications for distributed cache awareness.

## Tasks

COMPLETE THE FOLLOWING STEPS ON UBUNTU VM 2 AND WINDOWS VM 4

1. Locate the `cache.properties` file inside the web application directory:

   On Ubuntu VM 2 use the following command:

```
sudo nano /usr/local/jss/tomcat/webapps/ROOT/WEB-INF/classes/dal/cache.properties
```

   On Windows navigate to the path below then use Notepad++ to open the file.

```
C:\Program Files\JSS\Tomcat\webapps\ROOT\WEB-INF\classes\dal\cache.properties
```

2. Edit the file by modifying the `cache.type` value and delete the `DBAPPLIED_` string to overwrite previously applied settings in the database.

```
cache.type=memcached
```

3. Save the file after making changes.
4. Locate the `memcached.properties` file in the same directory as the `cache.properties` file.
5. Add the `memcached` endpoints enabled in Lesson 14 as follows:

```
memcached.endpoints[0]=192.168.56.102:11211
memcached.endpoints[1]=192.168.56.105:11211
```

   a. Uncomment the `memcached.uniqueId` attribute and modify the value:

```
memcached.uniqueId=ROOT
```

**NOTE:** Modifying the `memcached.uniqueId` attribute is required on Jamf Pro multi-context instances with memcached enabled.

6. Save the file after making changes.
7. Restart Tomcat.
8. The `memcached` service can be tested using `telnet` - see lesson 15 (step 6) for more details.

## End State

• All web applications in the cluster are configured for distributed caching

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# Load Balancing

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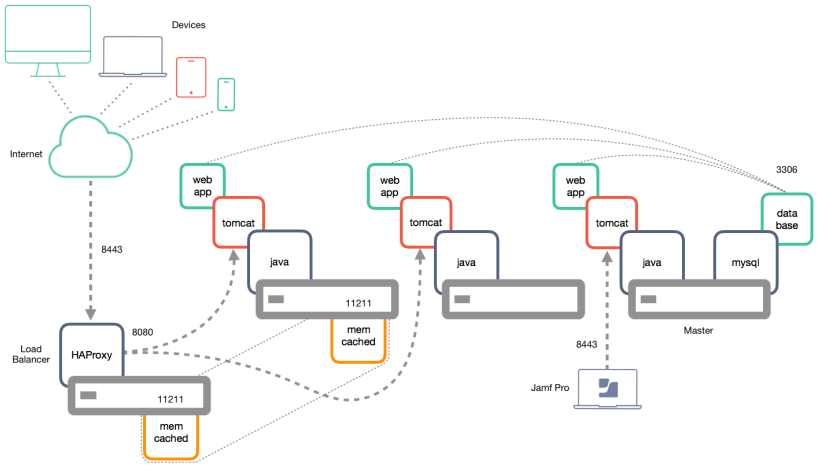
# Load Balancers




"Load balancing" is the distribution of workloads across computing resources. Load balancers improve performance by creating higher availability at backend servers and can also improve security.

• **Load balancer**: A reverse proxy server configured for load balancing, or, a dedicated load balancing hardware / software appliance.

• **Reverse proxy**: An intermediary server that distributes network traffic from an intranet or internet to other computers (typically a cluster of servers configured similarly to one another.)

• **Persistence / Session Affinity**: A way of ensuring a client stays connected through a load balancer to a specific web application in a cluster during a session.

• **SSL offloading / termination**: Interception of encrypted data by a server configured for a) decryption, and, b) the passing of decrypted data to other computers.

• **SSL acceleration**: SSL offloading, usually, on dedicated decryption co-processor hardware built into a load balancer appliance.

• **Proxy port**: The port to which a client initially directs its request (i.e., the port included in the URL) which may differ from the port serving the request. (e.g., `8443` vs. `8080`)

• **Proxy scheme**: The protocol scheme used for communication. (i.e., `http` or `https`)

• **X-forwarded-IP**: Ensures that `http` packet header information is maintained through a load balancer. (e.g., the originating IP address of a client computer)

• **RemoteIpValve**: A `valve` added to the Tomcat `server.xml` to enable X-forwarded IP.

• **Access log**: A Tomcat log that tracks user session activity on hosted web applications.

• **requestAttributesEnabled**: An attribute added to the `AccessLogValve` in the `server.xml` to enable passing IP address information to the Tomcat access log through a load balancer.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# Lesson 18: Load Balancer Installation and Configuration

## Objective

Install and configure HAProxy as a load balancer enabled for SSL termination on Ubuntu. Enroll the student computer into Jamf Pro.

## Discussion

• Master / child nodes
• Load balancer frontend / backend configuration

## Tasks

1. Open an `ssh` session to Ubuntu VM 5 in Terminal.app:

```
ssh serveradmin@192.168.56.105
```

COMPLETE THE FOLLOWING STEPS ON UBUNTU VM 5

2. Update the Ubuntu package repository index:

```
sudo apt update
```

3. Install HAProxy:

```
sudo apt install haproxy
```

4. Open the HAProxy configuration file in a text editor, e.g.,

```
sudo nano -Bc /etc/haproxy/haproxy.cfg
```

5. Add the text below to the `haproxy.cfg` file between the ciphers and the `defaults` label:

```
# Simultaneous connections the load balancer will allow
maxconn 2000

# Crypto Key size
tune.ssl.default-dh-param 2048
```

6. Modify the `haproxy.cfg` file by adding the following text below the `defaults` section:

```
# frontend definition
frontend tomcats
      bind *:8443 ssl crt /etc/haproxy/haproxy.pem
      option forwardfor
      reqadd X-Forwarded-Proto:\ https
      default_backend tomcats
```

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```
# backend definition
backend tomcats
      balance roundrobin
      cookie JSESSIONID prefix nocache
    # server webapp1 192.168.56.101:8080 check cookie webapp1
      server webapp2 192.168.56.102:8080 check cookie webapp2
    # server webapp3 192.168.56.103:8080 check cookie webapp3
      server webapp4 192.168.56.104:8080 check cookie webapp4
```

7. Execute the following command to verify the configuration:

```
sudo haproxy -f /etc/haproxy/haproxy.cfg -c
```

8. Copy the `haproxy.pem` file in the Student Resources directory from the Mac to Ubuntu VM 5:

```
scp .../haproxy.pem serveradmin@192.168.56.105:/tmp
```

9. Move the `haproxy.pem` to the location specified in the `haproxy.cfg` file:

```
sudo mv /tmp/haproxy.pem /etc/haproxy/
```

10. Restart HAProxy:

```
sudo service haproxy restart
```

11. Execute the command to verify the configuration:

```
sudo haproxy -f /etc/haproxy/haproxy.cfg -c
```

If the text in steps 5 and 6 was entered correctly into the `haproxy.cfg` file and the `haproxy.pem` was moved to `/etc/haproxy/` the test command should generate the following result:

```
Configuration file is valid
```

**Bonus!** Enroll the student computer:

a. Browse to `https://pro.jamf.training:8443/enroll`
b. Enter the credentials for the Jamf Pro user created with the Setup Assistant.
c. Skip the page to assign the computer to a different user by clicking Submit.
d. When prompted click Download to obtain the QuickAdd package.
e. Double-click the `.pkg` in the Downloads folder and follow the prompts to complete installation.
f. Browse to `https://pro.jamf.training:8443` to view the computer inventory record.
g. Note the reported IP address.

## End State

• A load balancer has been configured for the Jamf Pro cluster
• A signed SSL certificate has been installed on the load balancer to enable SSL termination
• The student computer has been enrolled to see the results of the X-Forwarded IP configuration


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# Lesson 19: Jamf Pro GUI Tomcat Load Balancer Settings

## Objective

Attempt to configure Tomcat for working behind a load balancer using the Jamf Pro GUI.

## Discussion

- Multi-context environment / Apache Tomcat Settings page
- Manual web application install
- Tomcat file permissions on Ubuntu: `chown`

## Instructor Demo

- Log in to Jamf Pro at `http://192.168.56.103:8080/context` with a web browser.
- In the top-right corner of the page click **Settings**.
- Click **Apache Tomcat Settings**, then, click **Edit**.

**NOTE:** on a mutli-context Jamf Pro server there is no Edit button on this page.

- Log in to Jamf Pro at `http://192.168.56.101:9006/` with a web browser.
- In the top-right corner of the page click **Settings**.
- Click **Apache Tomcat Settings**, then, click **Edit**.
- Select **Configure Tomcat for working behind a load balancer**, then, click **Next**.
- Select the **Enable Remote IP Valve** and **Enable Proxy Port** checkboxes.
- Verify that the **Proxy Port** is set to `8443`.
- Choose "HTTPS" from the **Scheme** pop-up menu, then, click **Next**.
- Click **Done**.

**NOTE:** the resulting error can be cleared by modifying ownership of the `server.xml` file

## End State

- The `server.xml` may (or may not) have been modified

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# Lesson 20: Manual Tomcat Load Balancer Configuration

## Objective

Configure Tomcat for working behind a load balancer by manually editing the `server.xml` file.

## Discussion

• Jamf Pro installer modified `server.xml`

## Tasks

COMPLETE THE FOLLOWING STEPS ON UBUNTU VM 2 AND WINDOWS VM 4

1. Locate the `server.xml` file.

2. Create a backup copy of the `server.xml` using any preferred method.

3. Verify that the `Connector` element for port `8080` is enabled. (Enable it by removing the surrounding comment tags. `<!-- -->`)

4. Add the `scheme` and `proxyPort` attributes to the `Connector` element as follows:

```
<Connector port="8080"
           protocol="HTTP/1.1"
           scheme="https"
           proxyPort="8443"
           redirectPort="8443"
           connectionTimeout="20000" />
```

5. Near the end of the file, above the final `</Host>` tag, add the `RemoteIpValve` element:

```
<Valve className="org.apache.catalina.valves.RemoteIpValve" />
```

6. Determine the Tomcat service name then restart Tomcat:

On Windows:

a. Hold CONTROL and click (or right-click) on the Start menu in the lower-left corner
b. Select "Task Manager" from the contextual menu.
c. Click on the "More details" button in the lower-left corner.
d. Click on the Services tab.
e. Hold CONTROL and click (or right-click) on the Tomcat service name.
f. Use the contextual menu to Stop, Start or Restart to restart Tomcat.


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On Ubuntu, execute the following command:

```
sudo service --status-all | grep -i tomcat
```

Use the discovered service name in a command like the following to restart Tomcat:

```
sudo service servicename restart
```

7. Browse to `https://pro.jamf.training:8443` to verify that the load balancer is intercepting and redirecting traffic for the web applications.

## End State

• Tomcat is manually configured for working behind a load balancer on the web applications in the cluster designated for handling client traffic

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# Lesson 21: Tomcat Access Logging

## Objective

Enable web applications in the cluster designated for handling client traffic to pass originating IP addresses through a load balancer to the Tomcat access log.

## Tasks

COMPLETE THE FOLLOWING STEPS ON UBUNTU VM 2 AND WINDOWS VM 4

1. Locate the `server.xml` file.

2. Create a backup copy of the `server.xml` using any preferred method.

3. Verify that the `AccessLogValve` element is enabled (enable it by removing the surrounding comment tags. `<!-- -->`)

4. Add the `requestAttributesEnabled` attribute to the `Valve` element as follows:

```
<Valve className="org.apache.catalina.valves.AccessLogValve"
       directory="logs"
       requestAttributesEnabled="true"
       prefix="localhost_access_log."
       suffix=".txt"
       pattern="%h %l %u %t &quot;%r&quot; %s %b"
       resolveHosts="false" />
```

5. Restart Tomcat.

**Bonus!** On the Mac execute the following command in Terminal.app:

```
sudo jamf recon
```

Browse to `https://pro.jamf.training:8443` to view the computer inventory record and note the reported IP address.

## End State

• The Tomcat access log is configured for receiving originating IP addresses through a load balancer


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# Security

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# Certificates

SSL certificates are critical for server security. Certificates validate website, web application and server host identity. Identity establishes trust allowing data to be encrypted in transit and decrypted for use.

- **SSL Certificates**
  1st party trust / 3rd party trust / Identity
  Public and private keys / Cryptography
  Keystores: `.jks`, `.pem`, `.pkcs12`, `.key`, `.der`, `.cer`
- **Certificate authorities**
  Internal / 3rd party
  Jamf Pro PKI Certificates
- **Web server certificates**
  Self-signed certificates
  Certificates signed by an untrusted issuer
  Certificates signed by a trusted issuer
- **Jamf Pro certificates**
  Certificate-based authentication
  Distribution points
  MDM Push Notification certificate
  MDM signing certificate
  SCEP device certificates
  jamf keychain
- **Tools**
  Keychain Access / `security`
  `keytool`
  `openssl`


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# Lesson 22: Keytool Introduction

## Objective

Use the Java `keytool` binary to create, view and delete a keystore file.

## Tasks

COMPLETE THE FOLLOWING STEPS ON UBUNTU VM 1

1. List the `keytool` command and subcommand options:

```
keytool
keytool -command -help [subcommands]
```

2. Generate a new public and private key pair by creating a keystore file. Follow the interactive prompts to create a password and the certificate distinguished name:

```
sudo keytool -genkeypair -alias test
```

3. View the keystore contents:

```
sudo keytool -list -v
```

4. View the `.keystore` file in its default path (the root user's home folder):

```
sudo su -
ls -lsa
```

5. Delete the keystore entry for the alias `test` created in step 2:

```
sudo keytool -delete -alias test
```

## End State

• A keystore file is generated and modified

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# Lesson 23: Ubuntu Certificate Keystore Creation

## Objective

Create a valid SSL certificate for Tomcat using Java `keytool` signed by the Jamf Pro Certificate Authority on Ubuntu.

## Tasks

1. Open an `ssh` session to Ubuntu VM 1 in the Terminal.app:

```
ssh serveradmin@192.168.56.101
```

COMPLETE THE FOLLOWING STEPS ON UBUNTU VM 1

2. Generate a new public and private key pair by creating a keystore file. Follow the interactive prompts to create a password and the certificate distinguished name:

```
sudo keytool -genkeypair -alias tomcat -keyalg RSA -keysize 2048 -validity 365 -keystore
/var/lib/tomcat8/keystore.jks
```

Use an integer for `validity` representing the number of days until certificate expiration.

3. Create a certificate signing request for generating a signed certificate from the Jamf Pro certificate authority. When prompted enter the password created above.

```
sudo keytool -certreq -alias tomcat -keyalg RSA  -keystore /var/lib/tomcat8/keystore.jks
```

The command output will appear in the Terminal.app window:


```
-----BEGIN NEW CERTIFICATE REQUEST-----
MIICxzCCAa8CAQAwUjELMAkGA1UEBhMCVVMxCzAJBgNVBAgTAk5ZMQwwCgYDVQQH
EwNOWUMxDDAKBgNVBAoTA2ZvbzEMMAoGA1UECxMDZm9vMQwwCgYDVQQDEwNmb28w
ggEiMA0GCSqGSIb3DQEBAQUAA4IBDwAwggEKAoIBAQD0Q/K7PUza56tNzTKkS511
WBEEv5mXcVtAQ6e18wKU6RnbiRABCKnGZnoLWFaqxVwQgjfHlHFjwQt5pis69Zg5
UDX4GsNhz5rH/+DjHx8je4r8XpF+bDFJYJ8pOUSf60eE5s4SRmevNdaxrBgwXQ4S
A4MPYQXv027DsDPSx77lnPcFcYDMOM5kUIh+XVaRYamQ6t8eudJstOuSlH6rIpkw
j5K2mwQZBJodMQut/ALQjieQsKS5G7Vab3b4pHDVrqBFw3HvUgUxT8t/WD0D+C9u
iYwOYSjFkYzp53C+1fg9XIvUj6LxEdynBuwpgpDSPVyBTb+n//yicl3uF8eHnT/H
AgMBAAGgMDAuBgkqhkiG9w0BCQ4xITAfMB0GA1UdDgQWBBSHdNcDSQuy6Xk56Hyu
ml/VR9s+jDANBgkqhkiG9w0BAQsFAAOCAQEALvVNATcjgB2goknQxEedSuiVc6cx
FqbFx2mT3yWemKfIi0ohdSXH9QF1OyN8+SJ+evLWk8syA7gNX/yw9lijkTA9NQ6W
1lr/sIDAV/k3/OizTCVhc2S6erdQ+iNf8zWvFUGCBQWfvHP96Cb4pGZmrq5URCiE
UtfOEPZidVhA5znf9T4Rcc5kzi3RiewP9i0nxOjHoQgql2RdVCtKz6BobGQCxDQk
p1Lx1G7HL8zFAtatgFn9VbbC9c5IQFvQ5m6QHY7rv3VmY833vgh+u24pPc3ulSBz
yxD91j9XSTg2DS/oxyqS7PyNH4bZsz3qYNf7NteM0OK/LdngPin9KHp3Mw==
-----END NEW CERTIFICATE REQUEST-----
```


4. Select all of the text generated by the command (including the `-----BEGIN NEW CERTIFICATE REQUEST-----` and `-----END NEW CERTIFICATE REQUEST-----` tags) and copy it to the clipboard using COMMAND + C or any preferred method. (The text can also be saved in a text file.)


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COMPLETE THE FOLLOWING STEPS ON THE MAC

5. Browse to `http://192.168.56.101:9006` to use the Jamf Pro certificate authority:

a. Enter the credentials for the Jamf Pro user created with the Setup Assistant to log in.
b. In the top-right corner of the page click **Settings**.
c. Click **PKI Certificates**.
d. Click on the **Manage Certificate Template** tab.
e. Click the **Download CA Certificate** button to download a copy of the Jamf Pro root certificate.
f. Click the **Create Certificate from CSR** button.
g. Paste all of the text generated in step 3 above into the **CSR** window.
h. Select **Web Server Certificate** from the **Certificate Type** pulldown.
i. Click **Create** to download the signed certificate.

6. Rename the `.pem` files in the Downloads folder:

a. Rename the Jamf Pro root certificate `root.pem` (The name will contain the string "JSS Built-in Certificate Authority".)
b. Rename the other downloaded certificate `reply.pem`

7. Copy the `.pem` files to Ubuntu VM 1:

```
scp ~/Downloads/*.pem serveradmin@192.168.56.101:/tmp/
```

COMPLETE THE FOLLOWING STEPS ON UBUNTU VM 1

8. Import the root certificate into the keystore. Note that the `-alias` used is `root`:

```
sudo keytool -import -trustcacerts -alias root -file /tmp/root.pem -keystore /var/lib
/tomcat8/keystore.jks
```

9. Import the signed certificate reply into the keystore. Note that the `-alias` used is `tomcat`:

```
sudo keytool -import -trustcacerts -alias tomcat -file /tmp/reply.pem -keystore /var/lib
/tomcat8/keystore.jks
```

10. Modify the `server.xml` to reference the keystore file:

a. Open the `server.xml` file with a text editor, e.g.,

```
sudo nano -Bc /var/lib/tomcat8/conf/server.xml
```

b. Enable the `Connector` element for port `8443` by removing the surrounding comment tags:

```
<!--
<Connector port="8443" protocol="org.apache.coyote.http11.Http11NioProtocol"
           maxThreads="150" SSLEnabled="true" scheme="https" secure="true"
           clientAuth="false" sslProtocol="TLS" />
-->
```

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c. Modify the `Connector` element for port `8443` to include the `keystoreFile` and `keystorePass` attributes:

```
<Connector port="8443"
           protocol="org.apache.coyote.http11.Http11NioProtocol"
           maxThreads="150"
           secure="true"
           clientAuth="false"
           scheme="https"
           SSLEnabled="true"
           sslProtocol="TLS"
           keystoreFile="/var/lib/tomcat8/keystore.jks"
           keystorePass="Jamf1234" />
```

The value of `keystoreFile` must be the absolute path to the keystore created in step 2. The value of `keystorePass` must be the password entered when the keystore was created in step 2.

11. Restart Tomcat.

## End State

- A valid SSL certificate signed by the Jamf Pro certificate authority has been created for Tomcat
- Tomcat is enabled for encrypted communication via `https` on port `8443`


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# Lesson 24: Windows Certificate Keystore Creation

## Objective

Create a valid SSL certificate for Tomcat using Java `keytool` signed by the Jamf Pro Certificate Authority. Configure access through the firewall for port 8443 on Windows.

## Tasks

COMPLETE THE FOLLOWING STEPS ON WINDOWS VM 3

1. Hold CONTROL and click (or right-click) on the Start menu in the lower-left corner, then, select "Command Prompt (Admin)" to open a new `cmd` window with elevated privileges.

2. Locate the `keytool.exe` file in the following location and drag the file into the `cmd` window opened in step 1, or, navigate to the `\bin\` directory using the `cd` command.

```
C:\Program Files\Java\jdk1.8.0_131\bin\keytool.exe
```

3. Generate a new public and private key pair by creating a keystore file.

To create the keystore password and certificate distinguished name on the Windows command line use the `-dname` `-keypass` and `-storepass` subcommands (as in the example below.)

To use the `keytool` interactive mode (as in the previous lesson) do not use the `-dname` `-keypass` or `-storepass` subcommands.

```
keytool.exe -genkeypair -alias tomcat -keyalg RSA -keysize 2048 -dname "CN=vm3.jamf.
training, OU=IT, O=Jamf, L=Minneapolis, ST=MN, C=US"
-validity 90 -keypass Jamf1234 -storepass Jamf1234 -keystore "C:\Program Files\Apache
Software Foundation\Tomcat 8.0\keystore.jks"
```

4. Create a certificate signing request for generating a signed certificate from the Jamf Pro certificate authority. This command will write a `.csr` file containing the public key to the Windows Desktop.

```
keytool.exe -certreq -alias tomcat -keyalg RSA -keypass Jamf1234 -storepass Jamf1234 -
keystore "C:\Program Files\Apache Software Foundation\Tomcat 8.0\keystore.jks" -file "C:
\Users\Administrator\Desktop\csr.csr"
```

5. Copy the `.csr` file to the Mac, open it with a text editor, select all of the text (including the `-----BEGIN NEW CERTIFICATE REQUEST-----` and `-----END NEW CERTIFICATE REQUEST-----` tags) and copy it to the clipboard using COMMAND + C or any preferred method.

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COMPLETE THE FOLLOWING STEPS ON THE MAC

6. Browse to `http://192.168.56.103:8080` to use the Jamf Pro certificate authority:

a. Enter the credentials for the Jamf Pro user created with the Setup Assistant to log in.
b. In the top-right corner of the page click **Settings**.
c. Click **PKI Certificates**.
d. Click on the **Manage Certificate Template** tab.
e. Click the **Download CA Certificate** button to download a copy of the Jamf Pro root certificate.
f. Click the **Create Certificate from CSR** button.
g. Paste all of the text generated in step 3 above into the **CSR** window.
h. Select **Web Server Certificate** from the **Certificate Type** pulldown.
i. Click **Create** to download the signed certificate.

7. Rename the `.pem` files in the Downloads folder:

a. Rename the Jamf Pro root certificate `root.pem` (The name will contain the string "JSS Built-in Certificate Authority".)
b. Rename the other downloaded certificate `reply.pem`

8. Copy the `.pem` files to the Windows VM 3 Desktop.

COMPLETE THE FOLLOWING STEPS ON WINDOWS VM 3

9. Import the root certificate into the keystore. Note that the `-alias` used is `root`:

```
keytool.exe -import -trustcacerts -alias root -file "C:\Users\Administrator\Desktop\root.
pem" -keystore "C:\Program Files\Apache Software Foundation\Tomcat 8.0\keystore.jks"
```

10. Import the signed certificate reply into the keystore. Note that the `-alias` used is `tomcat`:

```
keytool.exe -import -trustcacerts -alias tomcat -file "C:\Users\Administrator\Desktop\reply.
pem" -keystore "C:\Program Files\Apache Software Foundation\Tomcat 8.0\keystore.jks"
```

11. Modify the `server.xml` to reference the keystore file:

a. Open the `server.xml` file with Notepad ++. (Avoid using NotePad or WordPad.)
b. Enable the `Connector` element for port `8443` by removing the surrounding comment tags:

```
<!--
<Connector port="8443" protocol="org.apache.coyote.http11.Http11NioProtocol"
           maxThreads="150" SSLEnabled="true" scheme="https" secure="true"
           clientAuth="false" sslProtocol="TLS" />
-->
```


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c. Modify the `Connector` element for port `8443` to include the `keystoreFile` and `keystorePass` attributes:

```
<Connector port="8443"
           protocol="org.apache.coyote.http11.Http11NioProtocol"
           maxThreads="150"
           secure="true"
           clientAuth="false"
           scheme="https"
           SSLEnabled="true"
           sslProtocol="TLS"
keystoreFile="C:\Program Files\Apache Software Foundation\Tomcat 8.0\keystore.jks"
keystorePass="Jamf1234" />
```

The value of `keystoreFile` must be the absolute path to the keystore created in step 3. The value of `keystorePass` must be the password entered when the keystore was created in step 3.

12. Make a firewall exception for port `8443` to enable remote access to Tomcat by executing the following command on the Windows command line:

```
netsh advfirewall firewall add rule name=tomcat8443 dir=in action=allow protocol=TCP
localport=8443
```

13. Restart Tomcat.

## End State

- A valid SSL certificate signed by the Jamf Pro certificate authority has been created for Tomcat
- Tomcat is enabled for encrypted communication via `https` on port `8443`
- The Windows firewall has been configured for remote access on port `8443`

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# Practice Scenario 1

## Overview

Scenarios are not graded. They are intended to help set student expectations for the exam.

This workbook, course notes, text editors, text files, scripts, terminal applications, working Jamf Pro environments, Jamf Nation, the "internet", etc. can all be used as resources to complete the scenarios.

Asking the instructor questions and working with others is encouraged as long as those who wish to work alone or work quietly are respected. Using headphones for listening to audio is allowed.

## Objective

Build 2 load-balanced, clustered, multi-context Jamf Pro instances. Each cluster will have 2 nodes. A total of 4 web applications will be installed.

## Environment

| Name | IP Address | Hostname | Functions |
|---|---|---|---|
| Ubuntu VM 1 | `192.168.56.101` | `vm1.jamf.training` | web application<br>memcached |
| Ubuntu VM 2 | `192.168.56.102` | `vm2.jamf.training` | web application<br>memcached |
| Windows VM 4 | `192.168.56.104` | `vm4.jamf.training` | database |
| Ubuntu VM 5 | `192.168.56.105` | `pro.jamf.training` | load balancer |

## Preparation

- Locate the `350.vm.reset` file in the Student Resources directory on the Mac
- Double-click the file to execute the script
- Follow the instructions and prompts in the Terminal.app window
- Confirm with the instructor that execution is complete before starting the scenario


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COMPLETE THE FOLLOWING USING UBUNTU VM 1, UBUNTU VM 2 AND WINDOWS VM 4

| | Tasks | Lessons |
|---|---|---|
| ❑ | Configure Windows VM 4 as a database host. | 1, 10, 13 |
| ❑ | Use best practices for database naming, user names and passwords for MySQL database accounts. | 1, 10, 13 |
| ❑ | Configure Ubuntu VM 1 as a web application host. Allocate 512MB as the minimum and maximum memory amount for Tomcat Java Processes. | 6, 7 |
| ❑ | Install 2 unique contexts on Ubuntu VM 1: 1 named `alpha`, 1 named `bravo` | 11, 16 |
| ❑ | Configure Ubuntu VM 2 as a web application host. Allocate 512MB as the minimum and maximum memory amount for Tomcat Java Processes. | 6, 7 |
| ❑ | Install 2 unique contexts on Ubuntu VM 2: 1 named `alpha`, 1 named `bravo` | 11, 16 |
| ❑ | Set MySQL `max_connections` for the total number of web applications. | 13, 31 |
| ❑ | Configure logging for each web application. | 9, 11 |
| ❑ | Enable Clustering on both Jamf Pro instances and select one node in each cluster as the master. | 14, 15 |
| ❑ | Install memcached on Ubuntu VM 1 and Ubuntu VM 2. | 14 |
| ❑ | Configure memcached for multi-context Jamf Pro instances on all nodes in both clusters. | 15, 17 |
| ❑ | Configure Ubuntu VM 5 as a load balancer. All web applications must be configured behind the load balancer. | 18, 20 |
| ❑ | The load balancer must terminate SSL. Web applications behind the load balancer must be available via `https` and send traffic to Tomcat servers on port `9006`. | 8, 18 |
| ❑ | IP addresses of clients must report correctly to web applications and to Tomcat logs through the load balancer. | 20, 21 |

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# Lesson 25: Tomcat Security Configuration

## Objective

Enable strong security options for Tomcat.

## Discussion

- Disabling `http`
- Adding SSL protocols and ciphers
- Preventing server responses

## Tasks

COMPLETE THE FOLLOWING STEPS ON WINDOWS VM 3

1. Open the `server.xml` file in a text editor.

2. Locate the enabled `Connector` for `http` and disable it by placing comment tags as in the example below. The enabled `Connector` for `http` may be configured on a port other than `8080`:

```
<!--
<Connector port="8080"
           protocol="HTTP/1.1"
           redirectPort="8443"
           connectionTimeout="20000" />
-->
```

3. Modify the `Connector` element for port `8443` to match the following example by modifying the `sslProtocol` attribute and adding the `sslEnabledProtocols` attribute and its values:

```
<Connector port="8443"
           protocol="org.apache.coyote.http11.Http11Protocol"
           scheme="https"
           server="Apache"
           secure="true"
           clientAuth="false"
           SSLEnabled="true"
           sslProtocol="TLSv1.2"
           sslEnabledProtocols="TLSv1.2,TLSv1.1,TLSv1"
           keystoreFile="...keystore.jks"
           keystorePass="Jamf1234"
           ciphers="TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384,
                    TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384,
                    ...
```


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4. The list of ciphers to be added to the `ciphers` attribute can be:

- copied and pasted from a `server.xml` file created by the Jamf Pro installer, or,

- copied and pasted from the "Configuring Supported Ciphers for Tomcat HTTPS Connections" Jamf knowledge base article:

`https://www.jamf.com/jamf-nation/articles/384/configuring-supported-ciphers-for-tomcat-https-connections`

**NOTE:** A Jamf Pro server without the configurations set in steps 1-4 above is **NOT SECURE** unless it is configured behind a load balancer or proxy enabled for SSL termination, hosted on a secure perimeter network or firewalled on a closed, private network.

## End State

- Strong security options are enabled for Tomcat

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# Lesson 26: MySQL Security Configuration

## Objective

Execute the `mysql_secure_installation` script.

## Discussion

• MySQL: root access, account passwords, restricted grants
• mysql_secure_installation
• Tomcat / MySQL SSL Encryption

## Tasks

1. To execute the `mysql_secure_installation` script included with Windows MySQL server double click on then follow the interactive prompts.

```
C:\Program Files\MySQL\MySQL Server 5.7\bin\mysql_secure_installation.exe
```

2. To execute the `mysql_secure_installation` script included with Ubuntu MySQL server enter the following command at the shell prompt:

```
mysql_secure_installation
```

Follow the interactive prompts to complete.

If the command above fails to execute the script on Ubuntu, try the following:

```
cd /usr/bin/
sudo mysql_secure_installation
```

3. **OPTIONAL:** These steps could be performed manually achieving the same result as executing the `mysql_secure_installation` script:

a. Set a secure password for the MySQL root user.
b. Remove root accounts that are accessible from remote hosts.
c. Remove anonymous user accounts.
d. Remove the test database.
e. Ensure that only the MySQL root user can access the MySQL data directory.

## End State

• Strong security options are enabled for MySQL


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## Discussion: Restricted MySQL Grant Commands

• For a stand-alone web application or master node in a clustered environment the following MySQL privileges are required:

```
GRANT INSERT, SELECT, UPDATE, DELETE, CREATE, DROP, ALTER, INDEX, LOCK TABLES ON
databasename.* TO 'username'@'hostname' IDENTIFIED BY 'password';
```

• For a child node in a clustered environment the following MySQL privileges are required:

```
GRANT INSERT, SELECT, UPDATE, DELETE, LOCK TABLES ON databasename.* TO
'username'@'hostname' IDENTIFIED BY 'password';
```

MySQL privileges cannot be restricted by executing a restrictive grant command after "all" privileges have been granted. To set restricted privileges on a MySQL account it must be dropped and recreated, or, the privileges would have to be explicitly revoked with a command.

• To enable a MySQL user to view the processes of another MySQL user:

```
GRANT PROCESS ON *.* TO 'username'@'hostname' IDENTIFIED BY 'password';
```

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# Lesson 27: Enabling Limited Access

## Objective

Enable and understand Limited Access features for increased security on client-facing web applications in the Jamf Pro cluster.

## Discussion

- Limited Access features
- Customizing server responses for security
- Modifying the Jamf Pro MySQL database

## Tasks

1. Browse to `https://pro.jamf.training:8443/alpha/` or `https://pro.jamf.training:8443/bravo/` and log in. (If a clustered Jamf Pro instance is not available create one to practice configuring Limited Access.)

   a. In the top-right corner of the page click **Settings**.
   b. Click **Limited Access** then click **Edit**.
   c. Set all nodes in the cluster other than the master node to **Computer and Mobile Device Access**.
   d. **DO NOT** set `localhost` (i.e., the master node) to anything other than Full Access!
   e. Click **Save**.

2. Verify the configuration (which should take effect immediately) by browsing to the address of any Jamf Pro server on which Limited Access has been enabled. Browsing requests should be redirected to a white page saying "Web Application Disabled."

3. **OPTIONAL:** To modify the "Web Application Disabled" page open the following file inside the web application directory with a text editor:

```
.../WEB-INF/frontend/disabled.jsp
```

   a. After modifying the `.jsp` file, restart Tomcat.

If Limited Access has been unintentionally enabled it can be disabled by clearing the setting in the database.

- Open the MySQL CLI.
- Find the name of the Jamf Pro database to be modified:

```
SHOW DATABASES;
```


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• Execute the following command to read the Limited Access table in the Jamf Pro database:

```
SELECT * FROM databasename.limited_access_mode_settings;
```

The chart below shows the possible field values and their meanings:

| Value | Status | Devices |
|---|---|---|
| 0 | Limited Access disabled = log in enabled | - |
| 1 | Limited Access enabled = log in disabled | Computers Only |
| 2 | Limited Access enabled = log in disabled | Mobile Devices Only |
| 3 | Limited Access enabled = log in disabled | Computers and Mobile Devices |

• Reset Limited Access to its default setting by executing the following command:

```
UPDATE databasename.limited_access_mode_settings SET access_mode='0';
```

## End State

• Limited Access has been enabled on client-facing web applications

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# Lesson 28: Limited Access Configuration

## Objective

Limit access to User-Initiated Enrollment, the Jamf Pro API and the API documentation page.

## Discussion

- Tomcat servlets
- Jamf Pro authentication
- REST API

## Tasks

COMPLETE THE FOLLOWING STEPS ON UBUNTU VM 2

1. Stop Tomcat.

   On Ubuntu execute this command:

```
sudo service tomcat8 stop
```

   On Windows execute this command:

```
net stop tomcat
```

2. Locate the `web.xml` file in the Tomcat home directory, e.g.,

```
.../webapps/ROOT/WEB-INF/web.xml
```

3. Make a backup copy of the file.

4. Find the text `AccessFilter-Other` in the `web.xml` file.

```
<filter>
  <filter-name>AccessFilter-Other</filter-name>
  <filter-class>com.jamfsoftware.jss.frontend.JSSAccessFilter</filter-class>
   <init-param>
     <param-name>AccessMode</param-name>
     <param-value>Other</param-value>
   </init-param>
</filter>
```


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5. Locate the `<filter-mapping>` block directly below the `<filter>` block found in step 4. Add the `EnrollmentController` and `RestletServlet` lines inside the `<filter-mapping>` block directly above the closing `</filter-mapping>` tag matching the example below:

```
<filter-mapping>
 <filter-name>AccessFilter-Other</filter-name>
 <servlet-name>JSSCheckConnection</servlet-name>
 <servlet-name>PerformanceAutomationTriggers</servlet-name>
 <servlet-name>FileVault2InstitutionalKeyDownload</servlet-name>
 <servlet-name>FrontEndController</servlet-name>
 <servlet-name>FrontEndUploadController</servlet-name>
 <servlet-name>EnrollmentController</servlet-name>
 <servlet-name>RestletServlet</servlet-name>
</filter-mapping>
```

"Filtering" (i.e., disabling) servlets is contingent upon enabling Limited Access, e.g., if the `web.xml` on `192.168.56.102` is modified, Limited Access must be enabled on `192.168.56.102` on the Limited Access page in the Jamf Pro GUI as well for the configuration to take effect.

6. Locate the `api` directory inside the web application directory, e.g.,

```
.../webapps/ROOT/api/
```

7. Disable the `api` directory by setting its permissions to make it inaccessible, by moving it out of the web application directory or by deleting it.

8. Start Tomcat.

9. To verify the configuration:

   a. Browse to `https://pro.jamf.training:8443/enroll`
   b. Browse to `https://pro.jamf.training:8443/api`
   c. Execute the following command in a Terminal session on the Mac:

```
curl -k -u jamfadmin:Jamf1234 https://pro.jamf.training:8443/
```

## End State

- The User-Initated Enrollment page is unavailable on client-facing web applications
- The Jamf Pro API is unavailable on client-facing web applications
- The API documentation page is unavailable on client-facing web applications

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# Performance

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# Lesson 29: Database Management

## Objective

Configure and schedule automatic Jamf Pro database backups. Manually create Jamf Pro database backups using MySQL and Java commands.

## Discussion

- Log Flushing
- Database Maintenance
- JSS Database Utility / `JSSDatabaseUtil.jar`
- MySQL Commands: `mysqlcheck`, `mysqldump`, `mysql`
- MySQL replication

## Tasks

COMPLETE THE FOLLOWING STEPS ON WINDOWS VM 3

1. Copy the JSS Database Utility to the Windows Desktop from the Student Resources folder.

2. Double-click on the `JSSDatabaseUtil.jar` file.

3. Make the `JSSDatabaseUtil.jar` aware of the MySQL `bin` directory location if necessary.

4. Modify the information in the **Database Connection Setup** window to match the Windows VM 3 MySQL configuration in Lesson 1. (Alteratively, the JSS Database Utility could be connected to any currently functional, available Jamf Pro database.)

5. After connecting to MySQL test the functions of the JSS Database Utility by creating a back up, restoring a backup, scheduling a backup and scheduling backup deletions.

6. Hold CONTROL and click (or right-click) on the Start menu in the lower-left corner, then, select "Command Prompt (Admin)" to open a new `cmd` window with elevated privileges.

7. View the `JSSDatabaseUtil.jar` help menu for the `backup` subcommand by executing the following command:

```
java -jar C:\Users\Administrator\Desktop\JSSDatabaseUtil.jar help backup
```

8. Enter a command like the following to create a database backup file:

```
java -jar C:\filepath\JSSDatabaseUtil.jar backup -saveBackupTo C:\filepath -server
localhost -port 3306 -database jamfdb -user root -pass
```


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9. Restore a database backup with a command like the following:

```
java -jar C:\filepath\JSSDatabaseUtil.jar restore -file C:\filepath\jamfdb.sql.gz -server
localhost -port 3306 -database jamfdb -user root -pass
```

10. Schedule a backup with a command like the following:

```
java -jar C:\filepath\JSSDatabaseUtil.jar schedule -hour 2 -scheduledBackupPath C:\filepath
-deleteBackupsOlderThanDays 7 -server localhost -port 3306 -database jamfdb -user root -pass
```

11. The following commands can be used to verify MySQL data integrity. (If needed, navigate to the `bin` directory in the MySQL installation home directory to execute the MySQL binary commands.)

```
mysqlcheck
```

- Displays the help message and current variable values.

```
mysqlcheck -u root -p -1 jamfdb
```

- Lists all tables in the selected database and table status.

```
mysqlcheck -u root -p -A --auto-repair
```

- Checks database tables for corruption and attempts to repair it if found.

12. Create a backup of the database using the `mysqldump` command. When prompted enter the MySQL root user password.

```
mysqldump -u root -p -v --dump-date --lock-tables jamfdb > C:\filepath\jamfdb.sql
```

13. Restore any of the backup database files created above to a Jamf Pro database by using a command like the following:

```
mysql -u root -p -v jamfdb < C:\filepath\jamfdb.sql
```

**NOTE:** The `java` / `JSSDatabaseUtil.jar` and the `mysqldump` / `mysql` executables on Ubuntu use the same syntax as the commands above. Using these commands on Ubuntu requires using UNIX file paths.

## End State

- Automatic database backups are scheduled
- Database backup files are created using `JSSDatabaseUtil.jar` and MySQL commands
- A database backup file is manually restored

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# Lesson 30: Upgrading Scaled Environments

## Objective

Understand upgrading challenges. Automate the process of saving modified files inside the web application during Jamf Pro upgrades.

## Discussion

- Jamf Pro version upgrades
  - Clustered environment
  - Multi-context environment
- Jamf Pro component upgrades
- Debugging
- Migration

## Tasks

COMPLETE THE FOLLOWING STEPS ON WINDOWS VM 3

1. Stop Tomcat.

2. If not created in Lesson 29, create a backup copy of the `jamfdb` database.

3. Create a backup copy of the Tomcat home directory using any preferred method.

```
"C:\Program Files\Apache Software Foundation\Tomcat 8.0"
```

4. Delete the file named `ROOT.war` and the directory named `ROOT` in the Tomcat webapps directory.

5. To upgrade, copy the `ROOT.war` file from the Student Resources directory into the Tomcat webapps directory.

6. Restart Tomcat.

1. Copy the `maintain.jamf.pro.webapp.files.sh` and `ROOT.war` files from the Mac to the `/tmp/` directory on Ubuntu VM 1.

COMPLETE THE FOLLOWING STEPS ON UBUNTU VM 1

2. If possible, create a backup copy of the Jamf Pro database currently associated with Ubuntu VM 1.

3. Create a backup copy of the Tomcat home directory using any preferred method.


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4. View the `DataBase.xml` file in a text editor using the full path to the file. Note the `ServerName`, `DataBaseName`, `DataBaseUser` and `DataBasePassword` values.

```
sudo nano .../WEB-INF/xml/DataBase.xml
```

5. Execute the `maintain.jamf.pro.webapp.files.sh` script using the following command:

```
sudo sh /tmp/maintain.jamf.pro.webapp.files.sh
```

6. View the `DataBase.xml` again. Verify that `ServerName`, `DataBaseName`, `DataBaseUser` and `DataBasePassword` are not set to their default values. (The script moves the modified version of the file from the previously installed web application into the "upgraded" web application directory.)

## End State

- Backup copies of the databases and the web applications are created
- A web application is manually upgraded
- A web application is automatically upgraded with a custom script
- Modified files from the previous web application version are preserved after an upgrade

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# Lesson 31: Component Tuning

## Objective

Understand the settings for adjusting the number of connections created between Jamf Pro components.

## Discussion

- Jamf Pro `DataBase.xml`
  `MinPoolSize` / `MaxPoolSize`
  `show processlist;`
- My SQL `my.cnf` / `my.ini`
  `mysqladmin`
  `show variables like 'max_connections';`
- Tomcat `server.xml`
  `Executor`, `maxThreads`

### Jamf Pro default database connections

| `MinPoolSize` | `MaxPoolSize` |
|---|---|
| 5 | 45 |

The default `MinPoolSize` value ensures there are 5 connections between the web application and the database even if no data is actively being exchanged.

The default `MaxPoolSize` value sets a maximum of 45 connections between the web application and the database even if the number of connection requests exceeds this value.

### MySQL

| | **default** `max_connections` | **maximum** `max_connections` | **configuration file** |
|---|---|---|---|
| **Ubuntu** | 151 | 214 | `/etc/mysql/mysql.conf.d/mysqld.cnf > /etc/mysql/my.cnf` |
| **Windows** | 151 | ? | `C:\ProgramData\MySQL\MySQL Server 5.7\my.ini` |

The `max_connections` setting defines the total number of connections an instance of MySQL server will accept.


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The correct `max_connections` value for a Jamf Pro environment is directly based on the `MaxPoolSize` value and the total number of web applications deployed:

1 connected web application = (45 x 1) = 45 total possible connections
2 connected web applications = (45 x 2) = 90 total possible connections
3 connected web applications = (45 x 3) = 135 total possible connections
4 connected web applications = (45 x 4) = 180 total possible connections
etc.

If the total number of possible connections from all web applications connected to an instance of MySQL server exceeds the default `max_connections` value the `max_connections` value must be increased.

One (1) additional connection is often added to the total to ensure a thread is available for root user log in in the event of thread locks: (See Lesson 13 for step-by-step configuration instructions.)

To view the current setting for `max_connections`:

Windows - when prompted enter the MySQL root user password:

```
C:\Program Files\MySQL\MySQL Server 5.7\bin\mysqladmin.exe -u root -p variables | FINDSTR
max_connections
```

Ubuntu - when prompted enter the MySQL root user password:

```
sudo mysqladmin -u root -p variables | grep max_connections
```

## Tomcat

| | **default** `minSpareThreads` | **default** `maxThreads` |
|---|---|---|
| **Ubuntu** | 25 | 200 |
| **Windows** | 25 | 200 |

The `maxThreads` setting defines the number of connections between the web application and client devices (iOS devices, computers, etc.)

The suggested `maxThreads` value is loosely based on a ratio of 2.5 Tomcat threads (or connections) for each connection between the web application and MySQL. This value is calculated *per instance of Tomcat*.

1 installed web application = (45 x 2.5) x 1 = 113 threads
2 installed web applications = (45 x 2.5) x 2 = 225 threads
3 installed web applications = (45 x 2.5) x 3 = 338 threads
4 installed web applications = (45 x 2.5) x 4 = 450 threads
etc.

Because the default `MaxPoolSize` value in the `DataBase.xml` file has been lowered to 45 the Tomcat `maxThreads` value should only be increased in a multi-context environment, i.e., when using the ratio as a basis would result in a `maxThreads` value greater than the default value of 200.

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## Tasks

COMPLETE THE FOLLOWING STEPS ON WINDOWS VM 3

1. Locate and open the `DataBase.xml` file.

2. Lower the `MinPoolSize` value from 5 to 1.

```
<MinPoolSize>1</MinPoolSize>
<MaxPoolSize>45</MaxPoolSize>
```

3. Restart MySQL and restart Tomcat.

4. Execute the `show processlist;` command in the MySQL CLI to confirm the new `MinPoolSize` setting.

1. Locate and open the `server.xml` file.

2. Enable the `Executor` element by removing the comment tags:

```
<!--
<Executor name="tomcatThreadPool"
          namePrefix="Tomcat-"
          maxThreads="200"
          minSpareThreads="4"/>
-->
```

3. Set the `maxThreads` value for the number of web applications installed, e.g., (45 x 2.5) x 2 = 225

```
<Executor name="tomcatThreadPool"
          namePrefix="Tomcat-"
          maxThreads="225"
          minSpareThreads="4"/>
```

4. Add the `executor` attribute to all enabled `Connector` elements in the `server.xml` file using the example below as a guide. The `tomcatThreadPool` value functions as a variable which populates the `maxThreads` value set in the `Executor` element above in all connectors to which it is added.

```
<Connector port="8080"
           protocol="HTTP/1.1"
           URIEncoding="UTF-8"
           connectionTimeout="2000"
           redirectPort="8443"
           executor="tomcatThreadPool" />
```


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```
<Connector port="8443"
           protocol="HTTP/1.1"
           URIEncoding="UTF-8"
           clientAuth="false"
           scheme="https"
           executor="tomcatThreadPool"
```

5. Restart Tomcat.

## End State

- Relationships between Jamf Pro components and their settings are understood.
- The `DataBase.xml MinPoolSize` value is decreased.
- The `server.xml maxThreads` value is increased via the preferred method.

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# Practice Scenario 2

## Objective

Build 2 clustered, multi-context Jamf Pro instances for secure on-network (internal) and off-network (external / DMZ) device management. Each cluster will have 3 nodes with the master node in the cluster installed on the database server. A total of 6 web applications will be installed.

## Environment

Use any 3 of the 4 servers below:

| Name | IP Address | Hostname | Functions |
|---|---|---|---|
| Ubuntu VM 1 | `192.168.56.101` | `vm1.jamf.training` | |
| Ubuntu VM 2 | `192.168.56.102` | `vm2.jamf.training` | |
| Windows VM 3 | `192.168.56.103` | `vm3.jamf.training` | |
| Windows VM 4 | `192.168.56.104` | `vm4.jamf.training` | |

## Preparation

- Locate the `350.vm.reset` file in the Student Resources directory on the Mac
- Double-click the file to execute the script
- Follow the instructions and prompts in the Terminal.app window
- Confirm with the instructor that execution is complete before starting the scenario

## Limited Access Settings:

| **Limited Access** | GUI | API | Enrollment |
|---|---|---|---|
| External | DISABLED | DISABLED | DISABLED |
| Internal | DISABLED | DISABLED | **ENABLED** |
| Master | **ENABLED** | **ENABLED** | **ENABLED** |


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| | Tasks | Lessons |
|---|---|---|
| ❑ | Configure a server as a database host. | 1, 5, 10, 13 |
| ❑ | Use best practices for database configuration. | 1, 5, 10, 13 |
| ❑ | Enable strong security options for MySQL. | 26 |
| ❑ | The database server must also function as a web application host for the master nodes in the clusters. | 2, 3, 6, 7, 11, 13, 15 |
| ❑ | Configure a server as the internal web application host for child nodes in the clusters. Install 2 uniquely named contexts. | 2, 3, 6, 7, 11, 16 |
| ❑ | Configure a server as the external web application host for child nodes in the clusters. Install 2 uniquely named contexts. | 2, 3, 6, 7, 11, 16 |
| ❑ | Allocate 1024MB as the minimum and maximum memory amount for Java processes on all web application hosts. | 2, 6 |
| ❑ | Set MySQL `max_connections` for the total number of web applications. | 13, 31 |
| ❑ | Configure logging for each web application. | 9, 11 |
| ❑ | Enable Clustering on both Jamf Pro instances and select the node installed on the database server in each cluster as the master. | 14, 15 |
| ❑ | Install memcached on all Ubuntu servers. | 14 |
| ❑ | Configure memcached for multi-context Jamf Pro instances on all web application nodes in both clusters. | 15, 17 |
| ❑ | Set threads on each Tomcat server for the number of installed web applications using the preferred method. | 31 |
| ❑ | Tomcat on the internal and external web application servers must be configured for `https` with strong security options enabled and `http` disabled. | 23, 24, 25 |
| ❑ | Configure Limited Access on the internal and external web application servers using the "Limited Access Settings" on the previous page. | 27, 28 |

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# Lesson 32: Enabling Java Monitoring

## Objective

Enable the Java Management Extensions for monitoring Jamf Pro installations.

## Discussion

- Java Virtual Machine (JVM)
- Java Management Extensions (JMX)
- Managed Bean (MBean)
- Java Monitoring & Management Console (JConsole)

## Tasks

COMPLETE THESE STEPS ON UBUNTU

1. Make a backup copy of the `jmxremote.password` file with a command like the following:

```
sudo cp /etc/java-8-openjdk/management/jmxremote.password{,.bak}
```

2. Find the Tomcat user and group by executing the following commands:

```
cat /etc/passwd | grep tomcat
cat /etc/group | grep tomcat
```

3. Modify the ownership and permissions of the `jmxremote.password` file, e.g.,

```
sudo chown username:groupname /etc/java-8-openjdk/management/jmxremote.password
sudo chmod 400 /etc/java-8-openjdk/management/jmxremote.password
```

4. Open the `jmxremote.password` file with a text editor. Modify the `monitorRole` and `controlRole` lines at the bottom of the file by deleting the `#` symbols to match the example below:

```
...
# Following are two commented-out entries. The "measureRole" role has
# password "QED". The "controlRole" role has password "R&D".
monitorRole QED
controlRole R&D
```

5. Locate the `setenv.sh` file in the Student Resources directory on the Mac.

6. Copy the file from the Mac to the `/tmp/` directory of the Ubuntu server, then, move it to the Tomcat `/bin/` directory.


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7. Open the `setenv.sh` file with a text editor:

a. Set the `-Djava.rmi.server.hostname` value to match the Ubuntu server IP address.
b. Set the `-Dcom.sun.management.jmxremote.password.file` value to the absolute path to the `jmxremote.password` file.

8. Make the `setenv.sh` script executable. Use the full path to the file.

```
sudo chmod 755 .../bin/setenv.sh
```

9. Open the `DataBase.xml` with a text editor. Use the full path to the file.

```
sudo nano .../WEB-INF/xml/DataBase.xml
```

10. Modify the `DataBase.xml` by adding `<EnableJMX>true</EnableJMX>` immediately above the closing `</DataBase>` tag, e.g.,

```
<?xml version="1.0" encoding="UTF-8"?>
<DataBase>
...
     <EnableJMX>true</EnableJMX>
</DataBase>
```

11. Restart Tomcat.

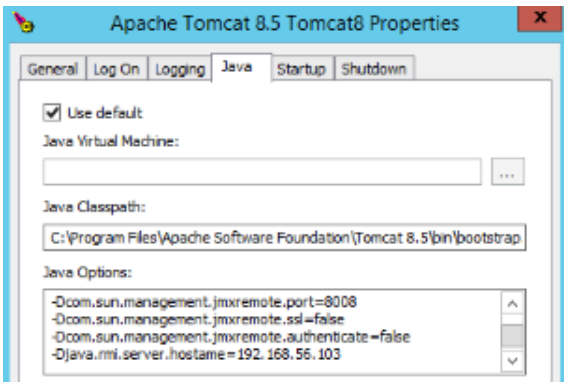
COMPLETE THE FOLLOWING STEPS ON WINDOWS

1. Use Notepad++ to open the `jmx.options.txt` file in the Student Resources directory.

2. Open the Tomcat Properties application located on the Notification Center Task Bar at the bottom right of the screen.

3. Select the Java tab.

4. The same security options (access, authentication, SSL, etc.) are available on Windows and Ubuntu, however, JMX can be accessed remotely without security. Use the `jmx.options.txt` file to populate the Java Options window on the Java tab of the Tomcat Properties application without changing any of the current settings as follows:

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5. Open the `DataBase.xml` file with Notepad++ and modify exactly as in the Ubuntu example above, then, save the changes.

6. Create a Windows firewall exception for port 8008 or set all firewall profiles to allow incoming connections for testing.

7. Restart Tomcat.

To verify that monitoring has been successfully enabled:

- Install the macOS Java JDK application from the Student Resources directory
- Open the JConsole application on the Mac by executing the following command in Terminal.app:

```
jconsole
```

When the JConsole window pops up select the "Remote Process" radio button, then, enter the following information:

- the IP address of the server and port in the suggested format (multiple simultaneous server connections are possible):

  Ubuntu port: 3333
  Windows port: 8008

- username: monitorRole (if authentication was enabled)
- password: QED (or whatever the password was changed to if changed)
- Ignore the "Secure Connection failed..." message by clicking Insecure Connection

## End State

- Detailed Jamf Pro server information is made available by enabling JMX monitoring
- Server activity is monitored via JConsole


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# Distribution Points

## Discussion

Methods of Distribution Point replication:

- Jamf Admin
- Jamf Distribution Server (JDS)
- JCDS / Cloud Distribution Points
- Private cloud / web-based / custom file sharing
- Manual copying
- `rsync` / file naming conventions
- Windows DFS / IIS
- box.com
- Resilio Sync (BitTorrent)
- Other solutions?

| | File Share Distribution Point | Cloud Distribution Point | JDS Instance |
|---|---|---|---|
| **Description** | Standard server that is configured to be a distribution point | Distribution point that uses one of the following cloud services to host files:<br>▪ Rackspace Cloud Files<br>▪ Amazon Web Services (S3 and CloudFront)<br>▪ Akamai | Distribution point that is managed by Jamf Pro, similar to a computer or mobile device |
| **Maximum Number** | Unlimited | One | Unlimited |
| **OS Requirements** | Any server with an Apple Filing Protocol (AFP) or Server Message Block (SMB) share | None | OS X or Linux (Server.app 2.2 or later required on OS X) |
| **Protocol** | AFP, SMB, HTTP, or HTTPS | HTTPS | HTTPS |
| **Authentication Options** | ▪ AFP or SMB:<br>  ▪ Username and password<br>▪ HTTP or HTTPS:<br>  ▪ No authentication<br>  ▪ Username and Password<br>  ▪ Certificate-based authentication | None | ▪ No authentication<br>▪ Certificate-based authentication |

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| | | | |
|---|---|---|---|
| **Files That Can Be Hosted** | ▪ Packages<br>▪ Scripts<br>**Note**: If You have "migrated" your scripts and packages or started past version 9 of Jamf Pro scripts are contained in the Database. | ▪ Packages<br>▪ In-house apps<br>▪ In-house eBooks<br>**Note**: If you use the cloud distribution point, scripts are stored in the jamfsoftware database. | ▪ Packages<br>▪ In-house apps<br>▪ In-house eBooks<br>**Note**: If you use one or more JDS instances, scripts are stored in the jamfsoftware database. |
| **Parent-Child Capabilities** | No | No | Yes |
| **File Replication Method** | Initiated from Jamf Admin | Initiated from Jamf Admin for Selective replication. If JDS is the Master distribution point and Cloud is set to Contents of Master the files will automatically replicate | ▪ Root JDS instance: Initiated from Jamf Admin if the Master is a file share. If the Master is a Cloud DP the replication is automatic.<br>▪ Non-root JDS instances: Automatic and immediate |
| **Selective File Replication** | Not available | Define the selective packages within the JSS. | ▪ Root JDS instance: Not available<br>▪ Non-root JDS instances: Available |


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